

GE Healthcare

MR - Signa[®] EXCITE[™] HD

Release Notes



Signa[®] EXCITE[™] HD
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Chapter 1

Purpose and Scope

Introduction

This document provides additional information that is not available in the current operator manual. This document, MR - Signa® EXCITE™ HD Release Notes Revision 9, replaces previous revisions. Please discard previous revisions and keep this revision with your MR system.

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Chapter 2

Scan

Introduction

This chapter has the following content:

- [PURE and other image filter considerations](#)
- [Graphic Rx](#)
- [Spectroscopy Graphic Rx](#)
- [Scan considerations](#)
- [SAR](#)
- [Manual prescan fat/water suppression](#)
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- [Equipment](#)
- [User CVs](#)
- [Bulk Susceptibility: fat SAT uniformity in C-spine](#)

PURE and other image filter considerations

Keep the following important points in mind when using PURE or other image filter options with scans that are post processed or displayed in a viewer - for example: Viewer, FuncTool, CADstream™, BrainWave, ReportCARD™, Flow Analysis™, StarMap™, etc.

- The PURE feature changes the pixel intensity in the image to provide a more uniform image. This may lead to altered results when CADstream is used in conjunction with PURE for processing MR images. A modification to the CADstream difference threshold may be required. It is recommended to either disable PURE for dynamic studies or applying the PURE feature uniformly across all the dynamic series.
- The output of post processing can be modified with any change to the protocol. For optimal image quality, keep consistency with protocols and use the same type of filter or apply SCIC or PURE for both pre and post contrast imaging.
- If using a third party post processing software, refer to their operator manual or contact your third party software representative regarding any change to your dynamic protocol and the use of PURE or other image filtering techniques.

Graphic Rx

Graphic Rx considerations

- If a Graphic Rx radial series is pasted to a Propeller series in the Rx Manager, no slices are posted and no error message is displayed. Radial prescriptions are not allowed for Propeller acquisitions.
- On occasion, in Graphic RX, the Calibration scan is automatically displayed instead of the 3-plane localizer or another valid series when a new series is in graphic prescription. To display a localizer series, click **Select Series** on the Graphic Rx window and select the 3-plane localizer or another series.
- The following message may appear when downloading a radial graphic prescription: **Error code during geometry calculation, represcribe slices/slab on Graphic Rx screen. Error code 2218682.** Click **New Series** from the Rx Manager, select the desired series from the Protocol Man window and click **Accept**. Prescribe the radial graphic prescription on this new series.
- Graphic prescription lines may appear not as prescribed. This happens as a result of the requirement that whenever a localizer set is changed after prescribing a graphic prescription or when automatic conversion from explicit to graphic Rx happens, the prescription needs to reset its center to the center of the localizer set.

Graphic Rx Automatic Launch

A site preference has been added that allows you to have one of the following states for Graphic Rx:

- Graphic Rx screen is automatically displayed when a series is placed into a View/Edit state. To view the scan desktop, click the **Close** icon, to close Graphic Rx.
- The scan desktop is automatically displayed when a series is placed into a View/Edit state. To view the Graphic Rx screen, click the **Graphic Rx** icon on the scan desktop.

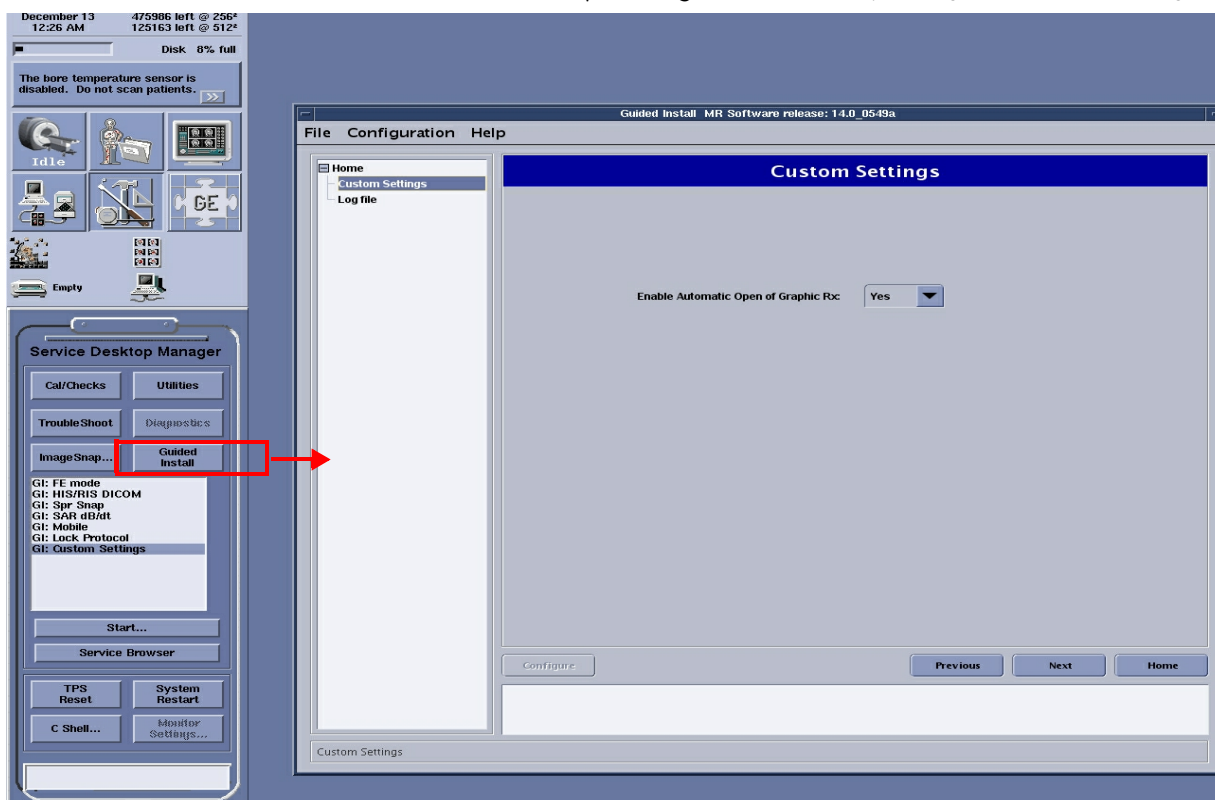
Considerations

- The Graphic Rx Automatic Launch mode applies to all PSDs.
- When the mode is set to On, the current rules of when Graphic Rx opens are maintained. For example, when the coil has changed the Graphic Rx window does not automatically open.
- When the mode is changed in the middle of an exam, a pop-up message informs that the new setting takes effect once a new exam is selected.

Procedure

1. Click the Service desktop icon.

Figure 2-1 Click **Guided Install** on the Service Desktop Manager. Service Desktop Manager and Custom Settings window.

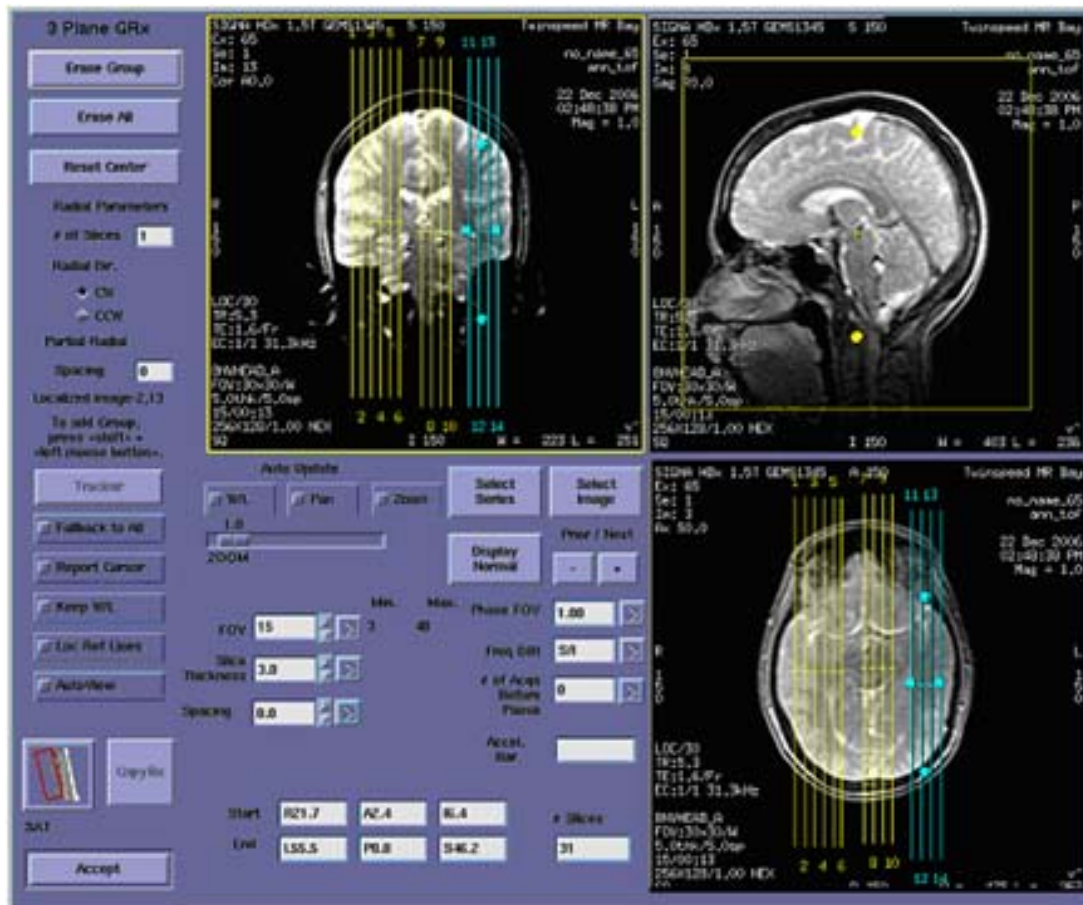


2. Double-click **Custom Settings** in the Service Desktop Manager list to display the Custom Settings window.
3. Make sure that Custom Settings is selected on the Guided Install window. Enable or disable Automatic Open for Graphic Rx from the pull-down menu (yes or no).
4. From the Custom Settings window, click **Configure** to accept your selection.
 - If an exam is open, the following message is displayed: **Please end exam and re-prescribe the patient details for the change to take effect.** Click **OK** to the prompt.
5. To close the Guided Install window, select **File > Close**.
6. To close Guided Install, select **File > Quit**. Click **Yes** to the confirmation prompt.

Graphic Rx workflow with Present Locs versus Original Locs

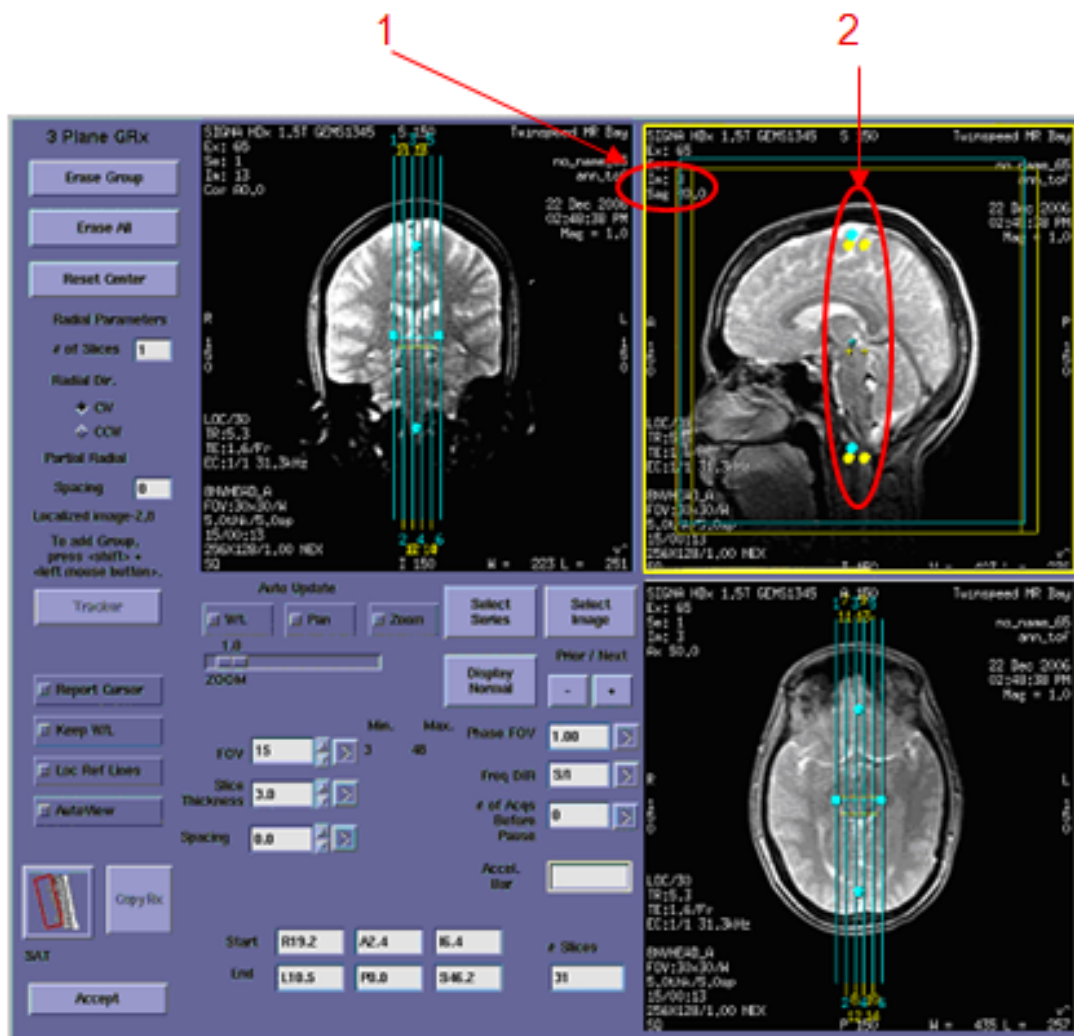
Graphic Rx workflow using Present Locs with multiple Grx groups.

1. You have graphically prescribed a multi-group acquisition, for example 3 sagittal locations on a coronal localizer. This is series 2 in this example.



2. You click New Series or another series in the Rx Manager and you want to prescribe the same set of slices. By default, the Sagittal image is automatically selected.
3. You want to copy/paste the graphic locations from series 2 onto series 3 so you click **Copy Rx** from the Graphic Rx panel and select **Present Locs**.
4. This prescription will result in all three sagittal images to be located at the same center value. As per the functionality provided for Present Locs option, all the slice groups re-localize to the center of the localizer. Thus, in this case, the center of all the groups get updated with the R/L value of the localizer center causing the visual effect of groups overlapping on top of each other.

Figure 2-2 1= Selected sagittal loc, 2 = Group centers have the same R/L value



To avoid this problem consider these rules when selecting Present Locs versus Original Locs from the Copy Rx List window:

Original Loc

Use this option when both the localizers with which the series was scanned AND graphic data (slices / 3D slabs / voxel) as placed in the series need to be copied.

Present Loc

Use this option when graphic data alone needs to be copied and **re-centered** on the new localizer that is highlighted in the graphic viewport.

5. To define the voxel thickness, change the value in the text box.
 - Do NOT click and drag the VOI to change the voxel thickness.
 - Do NOT use the **Start** and **End** Graphic Rx buttons to define the VOI for a 2D CSI scan.
6. Click **SAT** and deposit the SAT pulses.
7. Click **Graphic Rx** and click **Loc Ref Lines**. Verify that the VOI is bisected by the yellow reference line in the two planes that are orthogonal to the plane in which you deposited the VOI. In the example below, the VOI was deposited on the axial image. In image A the VOI size was only adjusted on the axial image. The VOI is bisected by the yellow lines in both the sagittal and coronal images. In image B, the VOI size was changed by clicking and dragging the VOI in the coronal plane. When the reference lines are displayed, it is clear that the VOI is not bisected by the yellow lines in the coronal and sagittal planes. The results are the following:
 - Image A will display the localizer when FuncTool is launched.
 - Image B will NOT display the localizer when FuncTool is launched.

Figure 2-4 Image A. Note that the VOI is bisected by the yellow cross reference lines.

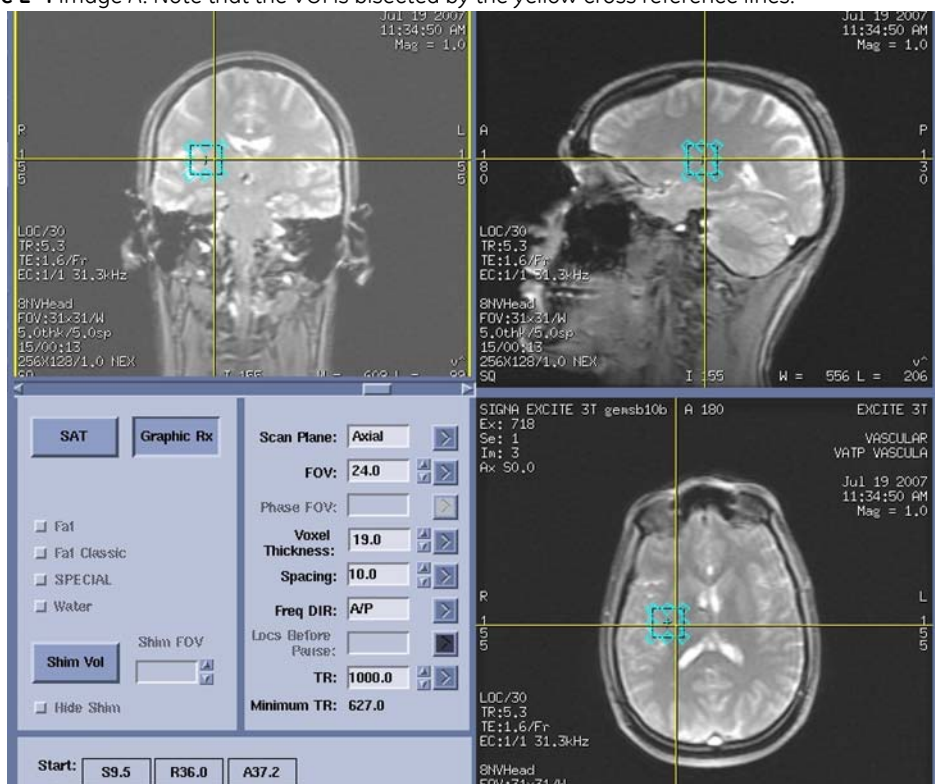
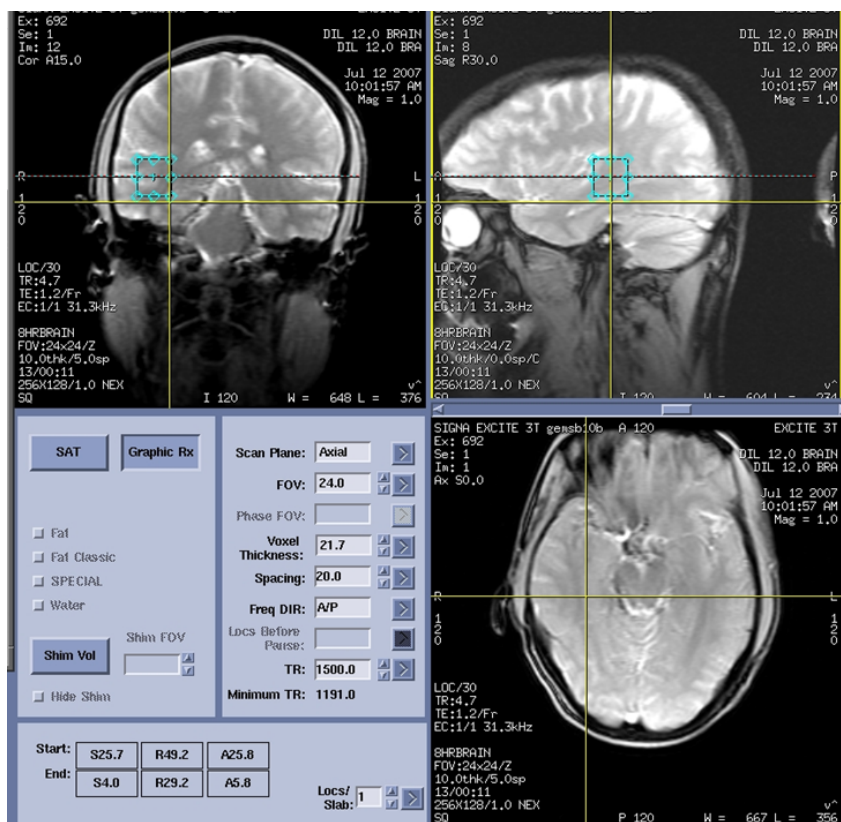


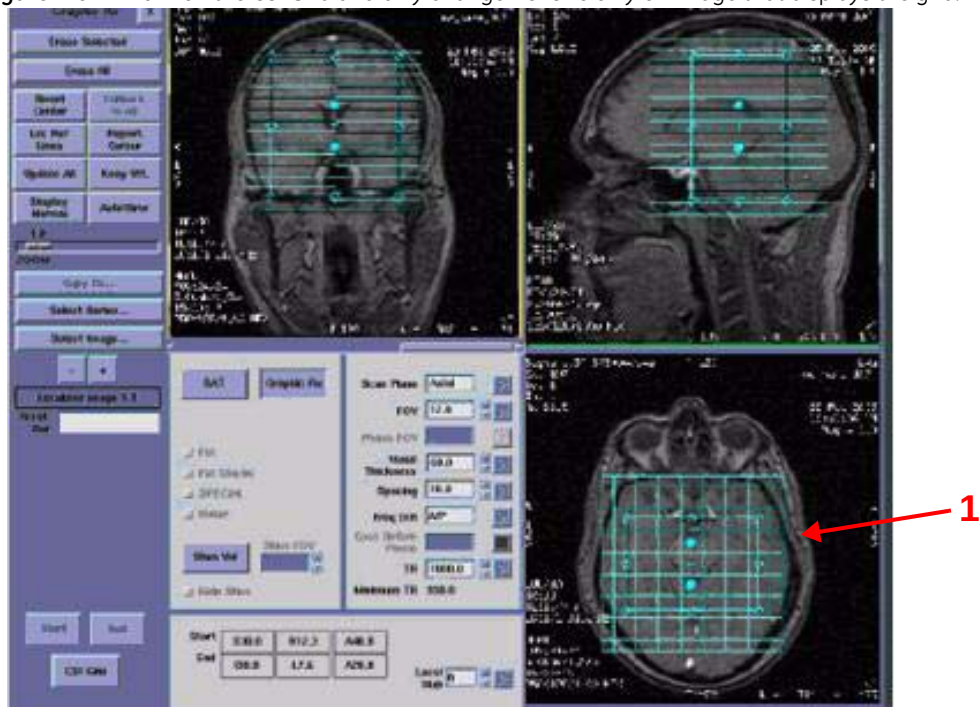
Figure 2-5 Image B. Note that the VOI is off center from the yellow cross reference lines. This prescription will **NOT** launch the localizer in FuncTool.



3D CSI prescription workflow

1. Place the sagittal and coronal images (if you are depositing the VOI on an axial) in two of the localizer viewports.
 - Use the middle mouse button to select the desired viewport. Using the left mouse to select these viewports will deposit the VOI and you are not ready to do that until step 3.
2. Place the image on which you want to deposit the VOI in the third viewport.
3. To define the first slice of the VOI, navigate to the desired slice and click **Start**.
4. Scroll to the last slice you want covered by the VOI and click **End**. This defines the VOI thickness.
 - Do NOT change the voxel thickness by any means after this step. If you need to change the VOI thickness, then click **Erase All** and start at step 3.
5. Click **CSI Grid** and only resize the VOI on the plane that displays the grid. In the following image, this is the axial image.

Figure 2-6 1 = Turn on the CSI Grid and only change VOI size only on image that displays the grid.



6. Click **SAT** and deposit the SAT pulses. Do NOT define the start and end VOI locations from the SAT page.
7. Click **Graphic Rx** and click **Loc Ref Lines**. Scroll to the image where you defined the start location. On one of the orthogonal images, verify that the yellow reference line passes through one edge of the VOI. Scroll to the slice where you defined the end location. On one of the orthogonal images, verify that the yellow reference line passes through the opposite edge of the VOI.
 - If the edges of the VOI do not align with the yellow reference lines, click **Erase All** and start over at step 3.

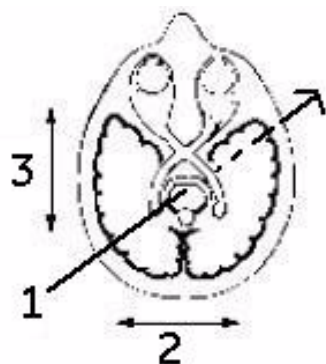
SAT Pulses in Spectroscopy

There are two scan interfaces from which you can prescribe SAT pulses on a spectroscopy acquisition: 6 from the Graphic RX SAT window, and 6 ROI Edge SAT pulses from User CV 18. The total number of SAT pulses that can be prescribed is 12. However, the maximum number that are actually applied is 10 or less than 10. The maximum number varies depending on the patient's weight and the selected coil.

If more than 10 SAT pulses are prescribed, all SAT pulses prescribed from the Graphic Rx interface are applied and one or more pair of ROI Edge SAT pulses is removed from User CV 18. The pair of SAT pulse removal occurs in the following order:

1. the pair that is parallel to the slice
2. the pair that is perpendicular to the phase direction
3. the pair that is perpendicular to the frequency direction

Figure 2-7 Order in which the VSS SAT pulses are removed when more than 10 SAT pulses have been prescribed.



Spectroscopy scan parameter considerations

There are three scan parameter fields that control if a spectroscopy scan is a single voxel, 2D CSI or 3D CSI acquisition:

- Frequency
- Phase
- # of CSI Slices
- The following table describes the value range for the three different prescriptions:

	Frequency	Phase	# of CSI Slices
Single voxel	1	1	1
2D CSI	> 1	> 1	1
3D CSI	> 1	> 1	> 1

It may be confusing when prescribing a single voxel spectroscopy scan because the frequency, phase and # of CSI Slices pull-down menus all display values greater than 1. The software was designed so that you can enter a value in all three fields on-the-fly to accommodate changing a prescription from SV to 2D or 3D CSI. It can be particularly confusing if you set frequency and phase values to 1 and then set the # of CSI Slices to a value greater than 1. The system does not allow this since it violates the rules defined in the table and therefore an error message is posted. Change the #of CSI Slices back to 1 and proceed with the prescription.

Localizer message

When you click **Save Series** for a 3D CSI PROSE or PROBE series in the Rx Manager, an Attention message box automatically displays.



Click **Continue** to close the message box. Re-prescribe the prescription so that the voxel thickness is a proper multiple of the localizer spacing. In this example, localizer spacing is 10, and the intended voxel thickness is 59.9, so voxel thickness needs to be changed from the incorrect value of 59.9 to the correct value of 60. This message box may also display after you click **View/Edit**. If this message is different than the one displayed when **Save Series** is clicked, it can be ignored.

Once the voxel thickness is a multiple of the localizer spacing, the following message displays.



Scan considerations

SCIC

SCIC is a filter applied to each 2D slice without using information from neighboring slices. This results in some signal intensity variation when images are reformatted. Therefore, do not use SCIC when acquiring scan data that is to be reformatted and use ClariView to apply SCIC as a post processing application. This workflow results in two data sets: non-SCIC from which you can reformat the images, and SCIC data for routine image viewing.

Calibration scan

An error message, “Part of scan region is outside the PURE/ASSET calibration volume” occurs when part of the area to be scanned is outside the area from the calibration scan. Be sure to scan the calibration at 48 cm FOV (or 30 cm FOV for the head) and cover the entire area of interest to alleviate this issue.

Patient Name characters

The patient’s name should be entered without numbers preceding the name. This text box is limited to 64 characters, although only 24 characters can be displayed on an image. If more than 24 characters are entered from the patient name field during scan prescription, then the displayed image shows the first 23 characters followed by *. The * indicates that more text was entered at scan prescription.

Prescan Time Change for 3.0T

On most 3.0T coils, the fast transmit gain process of Auto Prescan will not be activated. The result is improved image quality and a longer Auto Prescan time.

Manual Prescan

While 3-plane Graphic Rx is open and Manual Prescan is selected, the prescan window is in front of Graphic Rx window. If the MRSP option is selected, the MRSP screen pops up in front and the Manual Prescan screen goes behind all the other screens. This problem occurs when Modify RSP application is used from Manual Prescan window. Close 3-Plane Graphic Rx window to see the Manual Prescan window.

Saving Raw Data

Large raw data files (typically in excess of 100 MB) transferred via the network link and not the IT bus transfer at approx 3 to 4 MB/min. It can take as long as 20 to 30 minutes to transfer the raw data to hard drive. If the process is interrupted, the raw data files are incomplete and the final PXXXXX.7 files are not generated. Please be patient during the raw data transfer.

AutoViewer

AutoView quits updating if the system is in review mode and there are more than 124 1024x1024 images in the Auto View buffer. The circular buffer may get overwritten and result in images not available for review. To restart the AutoViewer:

1. Right-click in the background and click **Root Menu**.
2. Right-click **Restart Autoview**.

There are occasions when AutoView does not properly close once it has been opened on the MR Echo real time window. For example, when the real time window is closed in MR-Echo and the graphic window is opened, the AutoView window does not close. AutoView may also appear in the top right area of the window rather than the bottom right corner of the window. Toggle the AutoView button to close and open it.

New Patient screen is hidden

A white screen of 1024x1024 size is present on the Scan desktop is hiding the New Patient screen. This scenario can occur when the following steps are completed:

1. Graphic Rx is open on the Scan desktop.
2. Perform a TPS reset from the Tools desktop.
3. Remain on the Tools desktop until TPS reset is complete.
4. When the TPS reset is completed, switch to the Scan desktop.

Press **Alt+F3** to send white screen to the background.

Off-center FOV blurred images

Image blurring on 3.0T extremity images may occur with an off-center FOV and the zoom gradient coil. Use one of the following methods to eliminate this problem:

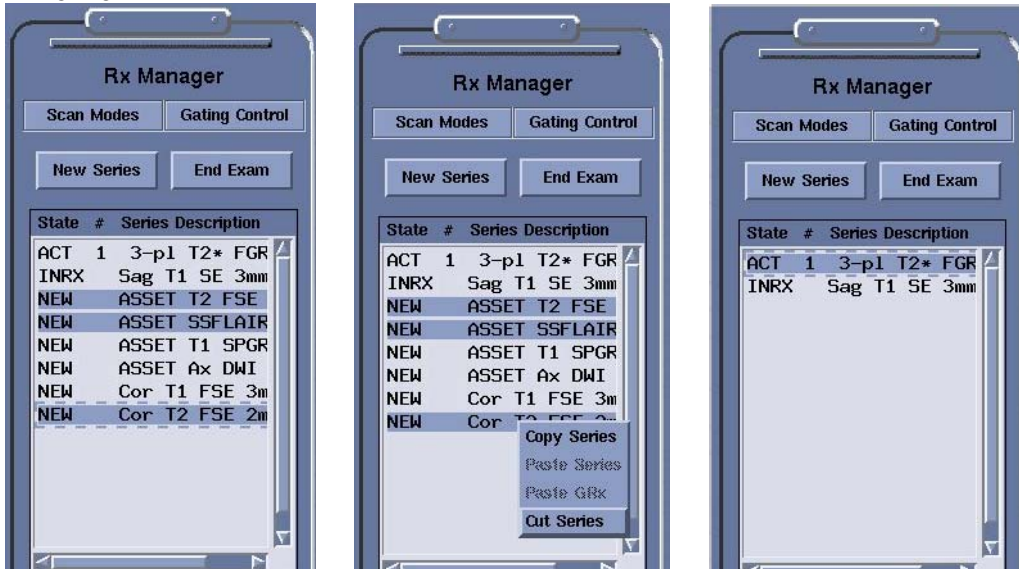
- Use Phase Correction with Zoom mode for off-center FOV imaging.
- Use Whole mode instead of Zoom mode for off-center FOV imaging.

SAT pulse thickness

If the default SAT pulse thickness of 40 mm does not effectively saturate signal, then change the SAT pulse thickness to 20 mm and use two SAT pulses adjacent to one another for 40 mm of coverage.

Cutting non-sequential series in Rx Manager

In the Rx Manager, if you highlight several series by pressing the **Ctrl** + series and the series are not consecutive, when you right-click **Cut**, all series from the first to last highlighted in the Rx Manager will be cut, rather than just the highlighted series.



To avoid this situation, highlight and cut one range of series at a time.

Apply/Apply All message

When you View/Edit a series in the Rx Manager and the coil selected for the series is different from that of prior series that have been prescribed, a message prompts you to make a selection of **Apply** or **Apply All**.



- **Apply** only changes the coil name on the series that is currently being viewed/edited.
- **Apply All** changes the coil name for all series in the Rx Manager below the currently view/edited series that are in a NEW or INRX state. Series above the currently view/edited series are not affected.



Workflow example

Rx Manager

Series 1 in ACT state with HEAD coil

Series 2 is inRX state with HEAD

Series 3 in RXD state with HEAD

Series 4 in NEW state with HEAD

Series 5 in NEW state with HEAD

1. In series 2, which is INRX state, change the coil to 8HRBRAIN.
2. Select **Apply All**. 8HRBRAIN is only applied to series 2, 4 and 5. Series 3 will not have the new coil applied since it is in an RXD state. The currently selected and connected coil is now the 8HRBRAIN coil.
3. View-edit series 3 which had HEAD coil as the saved coil selection. The Apply/Apply All message appears because the coil selected in series 3 is different than the previous series.
4. Click **Apply**. Series 3 now has HEAD as the currently selected and connected coil.
5. View/edit any of series 2, 4 or 5. The Apply /Apply All message box appears once again because the coil (8HRBRAIIN) is different from the previously selected coil in series 3.

NOTE: If you want to change the coil for all series within an exam, change the coil name on the first series and make sure that all series below it are in the NEW or INRX state. When you click **Apply All** the new coil is applied to All series in the Rx Manager and you will not get the pop-up message when you view/edit each subsequent series.

SAR



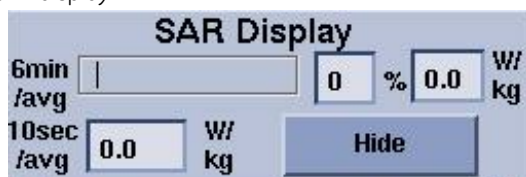
WARNING: The SAR algorithms for the 3.0T Signa system calculate SAR values and set a limit on the number of slices/echoes per second in order to limit RF power deposition. The power monitor and SAR algorithm limit SAR based on patient weight and RF transmit coil used. SAR limits are conservatively estimated from worst case patient positioning as a function of weight. The power monitor limits the RF power to limit the patient SAR to within the controlled limits over time.

SAR Display

The Specific Absorption Ratio (SAR) Display provides you with a visual representation of the average RF power delivered to the patient. The SAR Display shows the average RF power as the following:

- **6-minute interval:** top bar graph and numeric display
- **10-second average:** lower numeric display

Figure 2-8 SAR display



The 6-minute trip limit is 3 W/kg for head and extremity coils and 3 W/kg for body and receive only coils, based on the clinical FDA limit. The 10-second average regulatory limit is 3 times the 6-minute average limit. For example, while scanning an examination in the head coil, the scan aborts if the 6-minute average reaches 3 W/kg or the 10-second average reached 9 W/kg.

The bar graph indicates the cumulative RF power delivered to the patient for the 6-minute average across the examination, not just for each series.

SAR considerations

- Pulse sequence SAR predictions (estimated SAR) are based on patient weight at the worst-case landmark. To minimize nuisance power monitor trips caused by patient-to-patient variability, pulse sequence predicted SAR is the mean plus 1.96 standard deviations (typically the normalized standard deviation is about 18% at the worst-case landmark). The expected nuisance trip rate is around 2.5%. If you are seeing significantly more trips, please consult your local field service representative. The power monitor measures actual power and limits SAR appropriately.
- The power measuring accuracy of the power monitor is about +/-12%. Errors in patient weight do not result in excessive SAR. Low patient weight entries result in power monitor trips below the SAR limit. High patient weight entries result in fewer slices/images per unit time than would have been permissible.
- If the regulatory limit for SAR is reached, the scan pauses and you must wait until the 6-minute SAR average is within regulatory limits to complete the scan. The current series is lost or paused (depending on

the sequence), but all completed series are saved. In addition, you must wait until the SAR is within the regulatory limit before you are allowed to restart the series.

- The following scenario occurs with 3.0T Excite HD systems when SAR is operating in First Control mode. If the active exam is acquiring several SAR intensive series, such as SSFSE and FSE, before acquiring a PROPELLER series, then the long-term average SAR meter can exceed 2.0W/kg. This results in an SAR meter trip when you click **Download** for the PROPELLER series. To avoid the SAR meter exceeding 2.0W/kg, do one of the following:
 - Acquire the PROPELLER series after the localizer.
 - Change the acquisition order of the series in the protocol so that there is a time lapse (several minutes) between the SAR intensive series and the PROPELLER series.
 - Pause scanning several minutes before acquiring the PROPELLER series to allow the SAR meter's long-term average to fall below 50%.

First Level Operating Mode

- The Signa EXCITE 3.0T system uses the first level operating mode for SAR to determine the limits on scan parameters. The first time the prescription reaches the clinical mode limit of 2W/kg, a warning message appears. You must click either **[Accept]** or **[Cancel]**. If you click **[Cancel]**, the system prevents the scan prescription parameters from exceeding 2w/kg. If you click **[Accept]**, the system operates in the first level operating mode as defined by the IEC 60601-33 (Average whole body SAR limit > 2W/kg) for all series in the exam.
- Additionally, a First Level Operating Mode message appears. When the exam ends, the system resets to the normal clinical mode. When creating protocols, the system treats SAR levels the same as real-time prescriptions. If you want to always stay under 2W/kg, create your protocol by clicking **[Cancel]** when the warning message appears. If you click **[Accept]**, the system calculates the scan parameters based on the first level operating mode limits.
- For the 3T Whole Body with EXCITE technology, the first operating mode is not available and the system always operates in the 2W/kg mode.

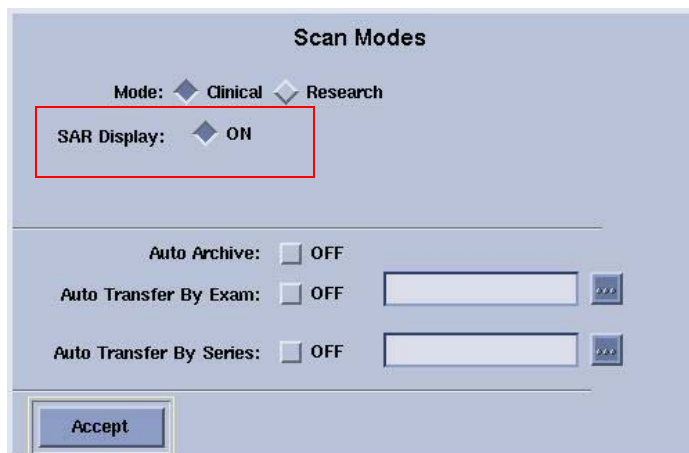
Improved pediatric SAR model

As the patient weight decreases, the number of slices based on the SAR may decrease. This may be more apparent when scanning pediatric patients.

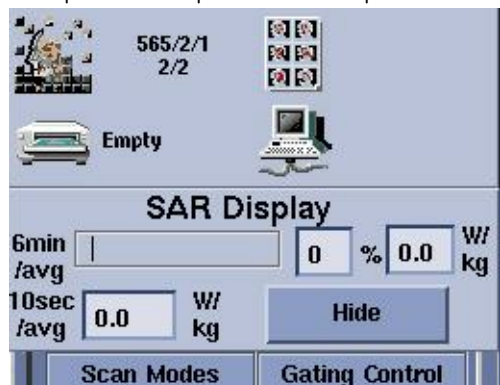
Scan workflow

To reduce the possibility of a power monitor trip due to high SAR values, interleave high SAR series such as FSE, FLAIR, and SSFSE with lower SAR series such as GRE and EPI.

1. Click **Scan Modes** and select the **SAR Display** toggle to **On**.



2. During the scan acquisition (not prescan), monitor the 10 sec SAR display. If the 10 sec SAR is > 3 W/kg, stop the scan and if possible represcribe the protocol with fewer number of slices or increase the TR.



- If you don't want to change the TR or the number of slices, then the following work-flow is recommended:
- Do NOT use auto-scan. Manually start each series when the 6 min average on the SAR display reads < 0.5 W.

Manual prescan fat/water suppression

Check and adjust CFA Fine for fat/water suppression

1. From the Manual Prescan window, click **[CF Fine]**.
2. Click **[Modify RSP]** from the Options pull-down menu.
3. To visualize the fat and water peaks, type and enter showfp in the RSP Name type-in field, and verify the Current Value is "1" for on, or showing the fat peak. Click **[Accept]**.
4. View the spectrum. If the desired peak is centered, skip to the next step. If not, adjust the Delta Freq value until the water or fat peak is centered over the vertical line. Use the fat peak as a reference point to confirm that the center frequency is set on water for fat suppression and fat for water suppression. The fat/water peak spread should be 220 +/- 20 Hz on 1.5T and 440 +/- 40 Hz on 3.0T. If all you want to do is check and adjust the center frequency, then when finished with this step click **[Done]** and proceed to scan. If you want to adjust the pulse to better suppress the fat/water peak, go to the next step.
5. Click **[Modify RSP]** from the Options pull-down menu.
6. Type and enter cstun in the RSP Name type-in field and change the current value from "0" to "1." Entering 1 turns on the presaturation pulse that is used for suppressing either fat or water. Click **[Accept]**.
7. Look at the spectrum to see if the peak you're trying to suppress is gone. If it no longer appears, you can leave the program. If a peak appears, type and enter csa in the RSP Name field.
8. CSA is a program that allows you to adjust the amplitude of the suppression pulse. Increasing the amplitude of the pulse may decrease the peak. Highlight the existing value, press **Delete**, and type in a new value 5-10 units higher or lower than the defaulted value. Click **[Accept]**. Repeat this step until you see the smallest fat/water peak.
9. Click **[Done]** when you are satisfied.

Protocols

Protocol Lockout

The Protocol Lockout feature protects protocols from being changed by unauthorized users. A case-sensitive password must be entered before you can save the protocol unless the protocol is being saved in the Other category. Contact your site administrator to gain protocol editing privileges or to get help accessing the protocol.

PSDs

FSE

Phase Correction

There have been changes made to Phase Correct that improve image quality (reduced edge blurring). These changes occur when Phase Correct is turned on (the new default state) for FSE and FRFSE pulse sequences. When Phase Correct is on, Auto Prescan may increase by 10-15 seconds.

Enhanced Fine Line Suppression, User CV 21

An Enhanced Fine Line Suppression technique for 2D FSE-XL acquisitions is now available through User CV 21. Fine Line artifacts are often seen in odd NEX scans and result from unsuppressed FID signals. There are two states for Enhanced Fine Line Suppression, User CV 21: 0 for off and 1 for on.

- User CV 21 = 0. This is the default state for the Fine Line Suppression, which results in the same usage seen in Ex HD M4 software and Ex 11.0 software. It applies gradients in a particular manner for all 2D FSE-XL prescriptions, is independent of the NEX selection, has a FOV and slice thickness cutoff (FOV $\leq$ 20 cm and slice thickness $\leq$ 6 mm), and 10% of the readout window is blanked out for true odd NEX scans.
- User CV 21 = 1. This is an enhanced version of Fine Line Suppression. This new version increases certain gradient pulses for true odd NEX scans thus improving the reduction of fine line artifact on odd NEX scans. When turned on with odd NEX scans, it can result in a longer echo spacing (no more than 10%) and thus a reduction of slices per TR. This becomes apparent when switching between odd and even NEX since these trade-offs are not seen with even NEX scans. If the User CV Blurring Cancellation is turned on, the new version of enhanced Fine Line Suppression will always be used, independent of the choice of NEX. In other words, there will be no changes to the echo spacing or number of slices per TR with NEX.

Figure 2-9 Odd NEX FSE-XL scans. Left (1)= no Fine Line Suppression technique, Right (2) = Enhanced Fine Line Suppression turned on.



3D FSE

When six locs per slab with Slices to Discard set to 1 are prescribed with a 3D FSE scan, some slice locations are duplicated. For example, slice one = location L26.5, slice two = location L25.5, slice three = location L24.5, slice four = location L23.5, slice five = location L24.5, slice six = location L23.5. To avoid this issue, select eight locs per slab for 3D FSE prescriptions.

3D FLAIR on 3.0T

When prescribing a 3D T2 FLAIR scan (3T only), you are restricted to the sagittal or sagittal oblique plane. Your Site protocols may need to be updated to reflect the necessary scan plane orientation.

Fast 2D Phase Contrast

If Flow Quantification Optimization is on, TE is not adjustable.

If the Flow Recon Type = Phase Diff, Flow Direction = In slice, and Flow Analysis = On and Flow Quantification Optimization is on, the flow encoding gradients are optimized and the TE is lengthened by approximately 1 ms.

FGRE- IR Prep

FGRE IR-Prep and PURE results in image artifacts due to PURE. The artifacts are more pronounced with the 8-channel cardiac coil. Turn off PURE with FGRE IR-Prep to avoid this issue.

Localizer considerations

- When selecting a PSD for a 3-plane localizer, if you first click **Fiesta** or **SSFSE** and then click **FGRE-IR Prep**, a pop-up message box may appear with the following text: **The IR Prep is not available with this pulse sequence.** Complete the following steps.
 1. Click **OK** to the message prompt.
 2. Click **Accept**.
 3. Click **New Series** from the Rx Manager.
 4. Prescribe your **FGRE-IR Prep** 3-plane localizer.
- A 3-plane localizer is the active prescription. If NPW and 1 NEX are both selected, the following incorrect error message is posted: **Please change the number of slices to 1.** The error message should read: **Please change the NEX to 2 or turn off NPW.** NPW and NEX = 1 are not compatible, so adjust your protocol and proceed to scan.
- If the patient experiences peripheral nerve stimulation from the localizer scan, to minimize the peripheral nerve stimulation probability on future three plane localizer scan, only use the FIESTA 3-Plane Localizer pulse sequence.

Spiral

You are prescribing a Spiral scan. If you delete the SSRF option from the Imaging Option field on the Imaging Parameters area of the scan desktop, rather than from deselecting it on the Imaging Parameters pop-up window, then the prescription user interface greys out (becomes invalid).

Figure 2-10 Do not delete SSRF from the Imaging Options text field.



Only delete an Imaging Option by deselecting the radio button from the Imaging Parameters pop-up window.

Spectroscopy

The spectrum may be inverted. Make sure the Imaging Option **EDR** is turned on. If it is on and the spectrum is still inverted, correct the inversion in FuncTool.

1. From the FuncTool desktop, click **Brain** or **Prostate** protocol.
2. Click **Advanced Settings** and then click the **Phasing** tab.
3. Click the Angle slider until the spectrum baseline becomes flatter or the peak is inverted.
4. Click **Compute** to recalculate the metabolite map and to display the new spectrum overlaid with the old spectrum in the upper right viewport.
 - Note that a message appears on the Reference metabolite map image indicating the phase shift adjustment: "Phase has been modified."

3D FSPGR neuro scans

The non-gated 3D FSPGR PSD has been modified to improve image quality. This modification may increase the minimum TR and decrease maximum number of slices per acquisition. This modification is applied to protocols when the following coils are used:

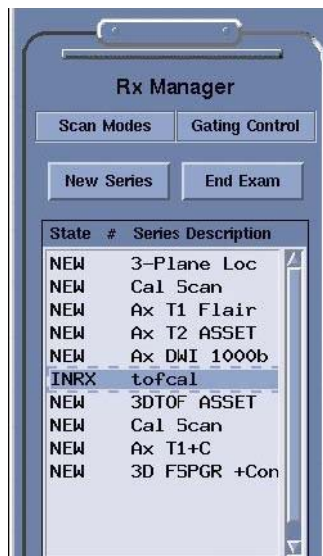
- Head
- 8HRBRAIN
- 8NV

TOFCAL

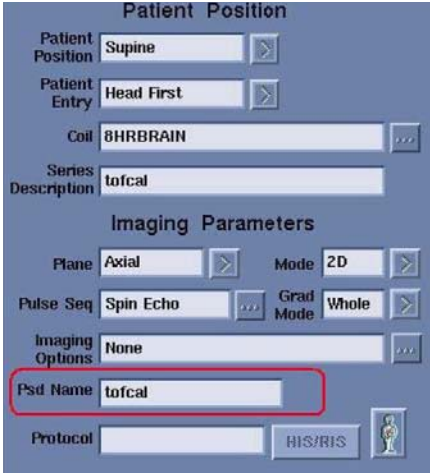
TOFCAL (Time of Flight Calibration) is a 3.0T application used exclusively when prescribing a 3D TOF scan with ASSET in conjunction with the 8HRBRAIN coil. This enhanced calibration is designed to improve coil sensitivity and to avoid black dot artifacts sometimes seen on 3D TOF in conjunction with ASSET for intracranial vasculature assessment.

TOFCAL workflow

1. Because the TOFCAL must be prescribed after a standard Calibration Scan an optimized workflow is shown below.



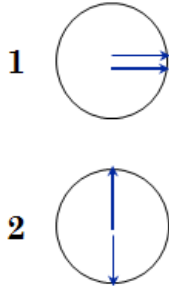
2. After a 3-Plane Localizer scan, prescribe graphically a Calibration Scan, and run any routine ASSET scans.
3. Prescribe the new TOFCAL scan just prior to 3D TOF ASSET. If additional ASSET scans are required (i.e. post contrast series), then a new standard Cal Scan must be prescribed prior to any other ASSET scans.
 - TOFCAL protocol.

TOFCAL Protocol	
Protocol	Select the TOFCAL protocol under GE protocols/other/tofcal_3T_12x. If the protocol is not in the library, then build the TOFCAL file and save it to your Site library.
Imaging Parameters	 Plane = Axial, Imaging Mode = 2D, Pulse Seq = Spin Echo, Imaging Options = None. On TwinSpeed systems, Gradient = Whole. In PSD name window type tofcal.
Scan Time	TR = 725, TE = 8.5, flip angle = 90°, Bandwidth = 31KHz. These parameters are optimized to give the highest SNR in the shortest scan time to collect the coil sensitivity images and therefore cannot be changed.
Scan Range	FOV = 30, slice thickness = 4, spacing= 4 These parameters cannot be changed.
Acq. Time	Phase and Frequency matrix= 32x32, NEX = 1 and cannot be changed. Frequency Direction = R/L, Shim =Auto
Graphic	Prescribe 26 slices only to cover the whole head; this is about 20cm coverage in S-I direction.

Gradient Echo Chemical Shift

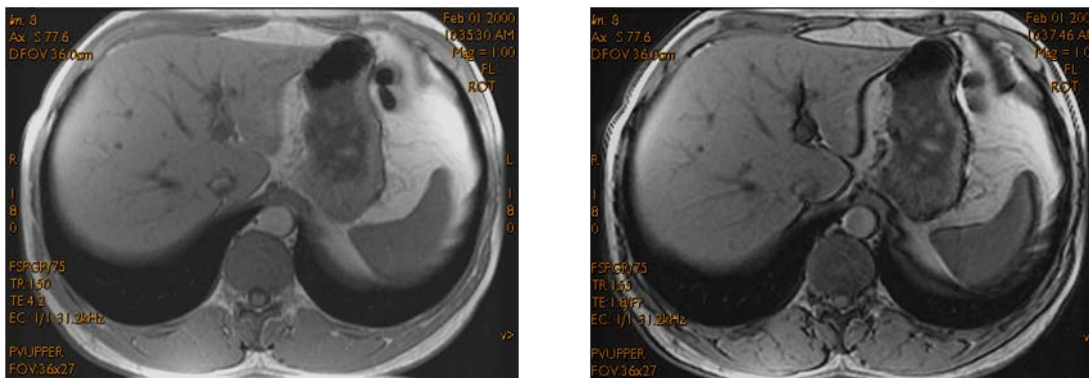
Chemical shift effects are seen when a voxel contains both fat and water and the TE is timed for the vectors to be in or out of phase.

Figure 2-11 1 = vectors in phase and 2 = vectors out of phase



Boundaries between fat and tissues with much water are either bright or dark. For some PSDs, such as Fast Spoiled Gradient Echo (FSPGR) the TE can be selected as In Phase or Out of Phase. When In Phase is selected, the fat/water border is bright and when Out of Phase is selected, the fat/water boundary is dark and structures can appear as if they have been outlined with a black marker.

Figure 2-12 Left = fat/water in phase, right = fat/water out of phase



To minimize the black border outline effect, select In Phase as the TE parameter, or manually enter a TE parameter as close to the fat/water in phase time as possible.

TE is in phase at time TE =0

	1.0T TE	1.5T TE	3.0T TE
Out of Phase	3.4 ms	2.3 ms	1.1
In Phase	6.8 ms	4.5 ms	2.3
Out of Phase	10.2 ms	6.8 ms	3.4
In Phase	13.6	9.1	4.5

PROPELLER

When acquiring a T2 FLAIR PROPELLER scan on 3.0T and phase wrap appears on the image, a subtle dark ring artifact may appear in the anatomy. To avoid this issue, scan with a larger FOV or center the anatomy exactly in the center of the FOV to eliminate the phase wrap which will eliminate the ring artifact.

PROPELLER artifacts

This section describes artifacts that are either unique or are manifested differently when using PROPELLER. Each is illustrated below along with suggestions to minimize the artifact.

Consider the following tradeoffs when prescribing PROPELLER:

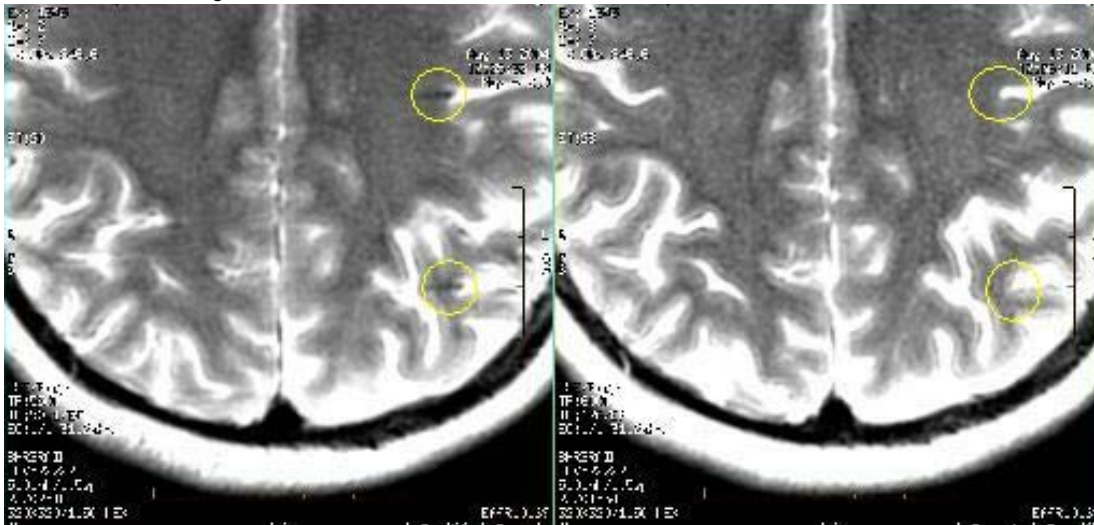
- As frequency resolution increases, the chance of crinkling increases; increase ETL to reduce crinkling
- As ETL increases, TE increases...increase Bandwidth to lower TE
- As Bandwidth increases, SNR decreases

Crinkling

The crinkling on PROPELLER acquisitions can be caused by:

- The wrong ETL. The wider the blade (determined by ETL), the more data for blade comparison and correction. This may appear in the superior slices. The most common cause is low ETL prescriptions and can be alleviated by increasing the ETL.

Figure 2-13 Left: ETL=20 (TE=90), Right: ETL=28 (TE=126)

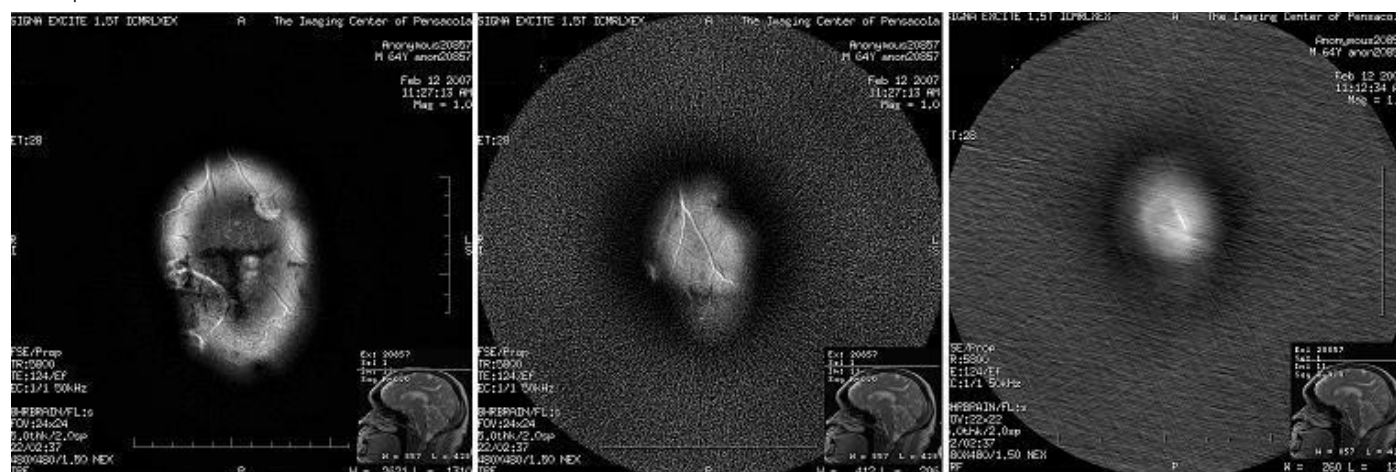


- Extreme patient movement. Patient movement during any acquisition will result in sub-optimal images. In PROPELLER, the image may be captured at an angle. Increasing the ETL may result in better motion reduction, but as with all motion-induced artifacts, eliminating patient movement is the best solution.
- Phase Wrap. Be sure the FOV fully contains the anatomy.

Top of scalp artifacts

Top of scalp artifacts are caused by not having enough data to perform motion correction. Prescribe fewer slices to cover only anatomy of interest.

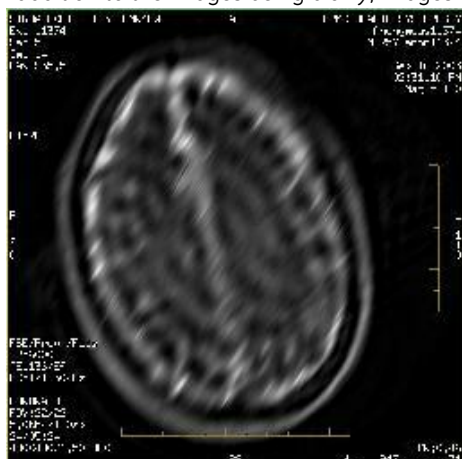
Figure 2-14 Left = Top of scalp; Middle = Extreme top of scalp displaying white pixels; Right = Extreme top of scalp displaying noise and white pixels.



Blurred Images

Blurred images can be seen with all 3 Propeller applications and results from a failure to match enough blades. If the acquired blades correlate poorly with each other, the following message appears in the GE syslog: **Image quality may be degraded due to poor blade correlation...**

Figure 2-15 In addition to the images being blurry, images may be captured at an angle.

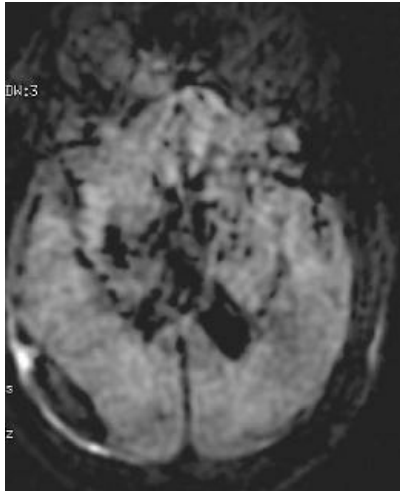


Signal Voids

Signal Voids can appear in diffusion weighted images produced by DW PROPELLER and are often indicated by black holes, but it may not occur with every scan.

Signal Voids may be caused by system calibrations out of spec (e.g. eddy currents). Contact service to ensure the system is within specification.

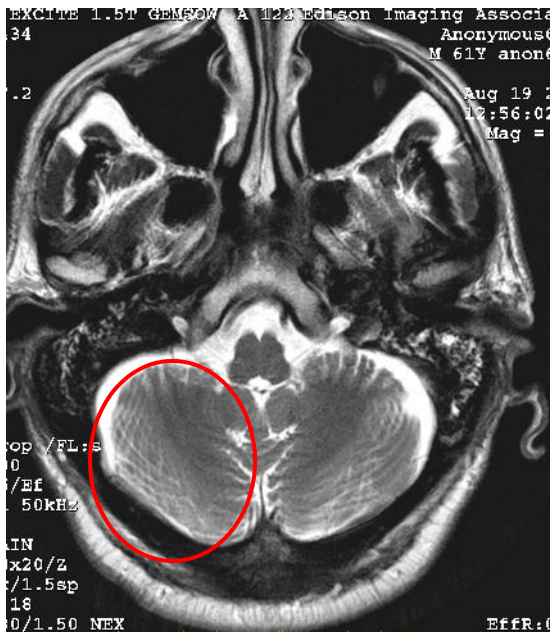
Figure 2-16 Signal void



Ripples

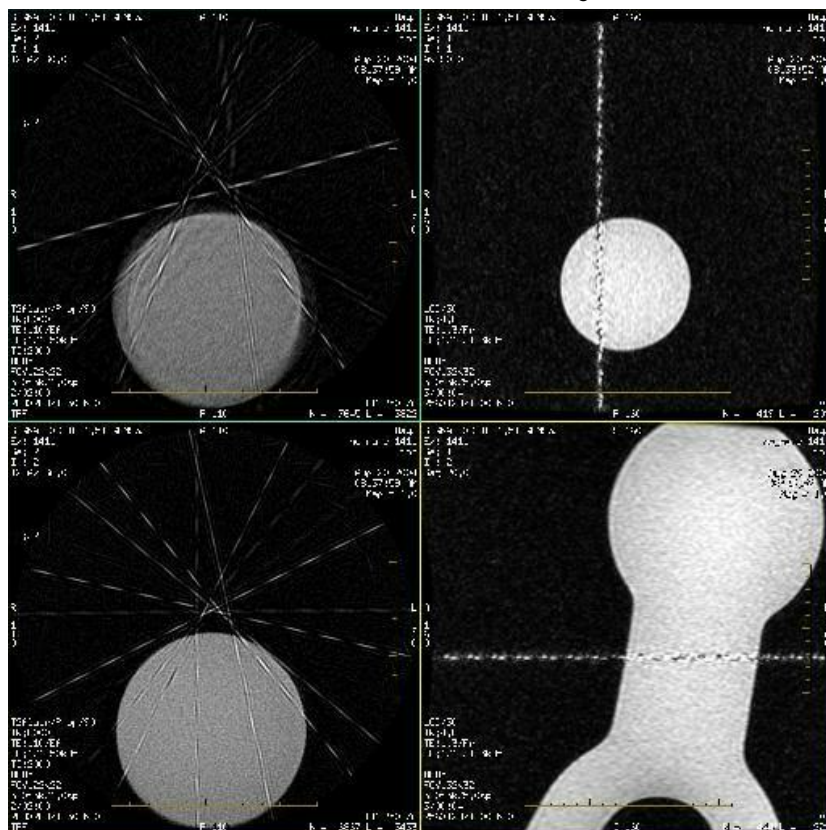
Ripples can be caused by wrap and can be seen with all 3 PROPELLER applications, but looks different due to the radial acquisition. To avoid wrap, increase the FOV and/or reposition the slice prescription.

Figure 2-17 Signal from the nose wraps around as radial lines or ridges



RF leak

Figure 2-18 Left = PROPELLER shows RF leak as criss-cross lines. Right = GRE shows RF leak as zipper artifact.



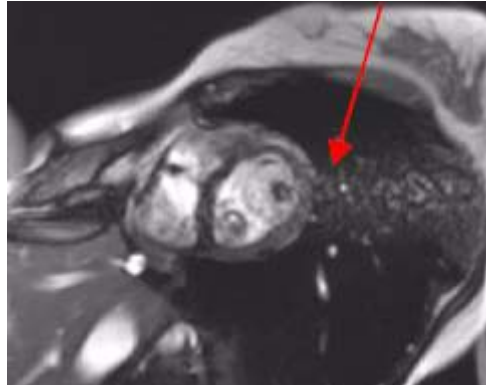
T2D Gated FIESTA Enhancement

The GE Signa® Excite HD 1.5T and 3.0T MRI systems have incorporated a new prescan and shimming technique to improve gated 2D FIESTA sequences that are prescribed in a single slice group.

Application

2D gated FIESTA is primarily used in cardiac imaging to assess cardiac function. Cardiac FIESTA imaging often results in off-resonance artifacts when the center frequency is incorrectly obtained. These artifacts appear as inhomogeneous areas within the blood pool and as ghosting across the image. They are more prominent in areas of fast or turbulent blood flow.

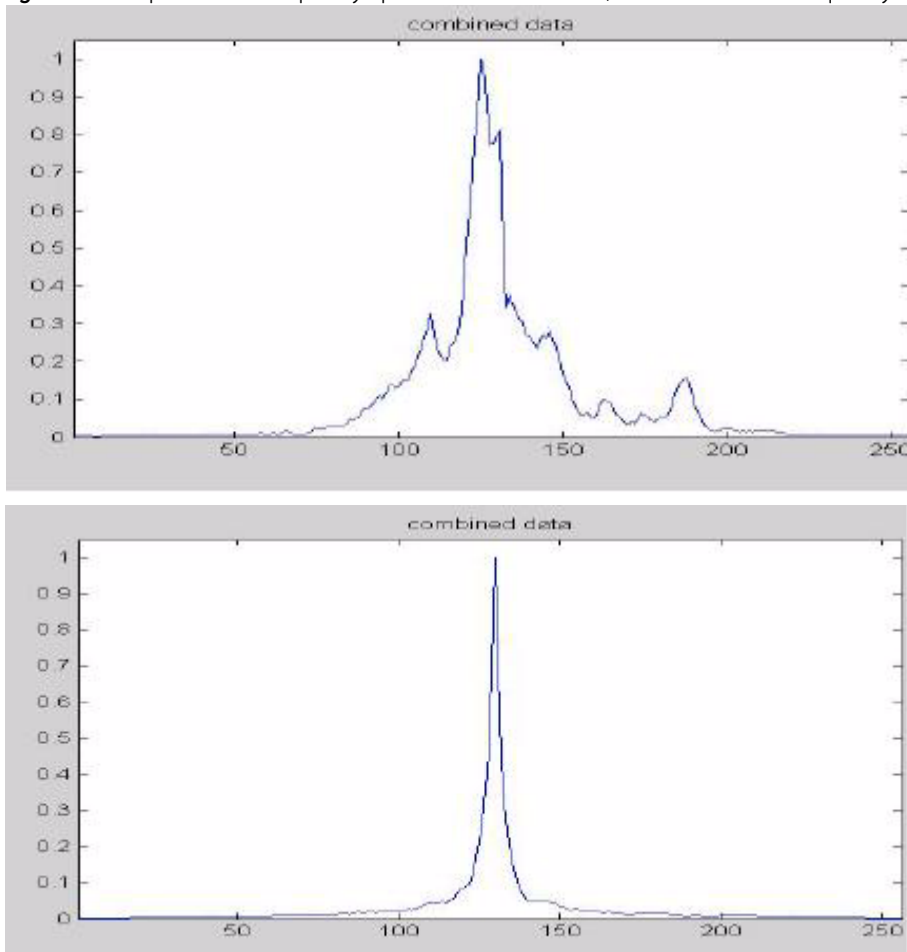
Figure 2-19 Cardiac image with off-resonance effects.



The enhanced FIESTA sequence significantly reduces off-resonance effects that are sometimes encountered when the correct center frequency of a cardiac image is difficult to obtain. The following enhancements in the 2D FIESTA prescan technique result in a significantly improved spectrum:

- Suppression of fat signal
- Only displaying the frequency spectrum of the relevant FOV

Figure 2-20 Top = Center Frequency spectrum for 2D FIESTA, Bottom = Center Frequency spectrum for enhanced FIESTA sequence.



Procedure

2D gated FIESTA Graphic Prescription

1. Place a small FOV shim volume over the heart. The shim volume FOV should be approximately half the size of the scan FOV. For example, if the scan FOV is 36 cm, then the shim volume FOV should be 18 cm.

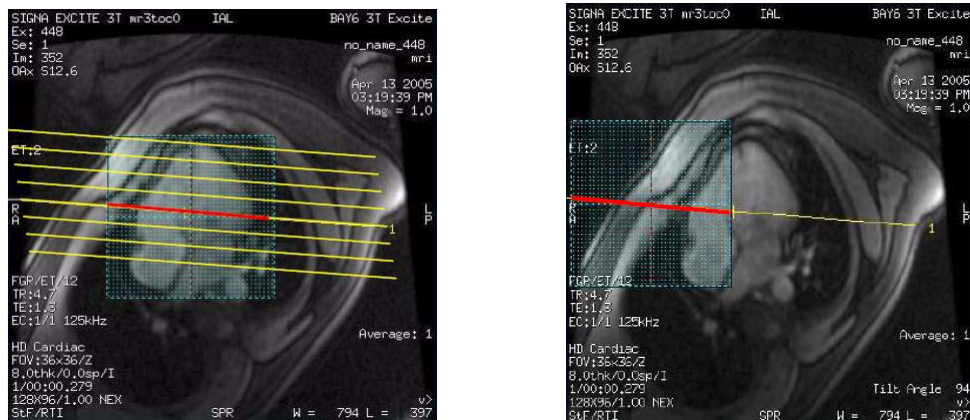
Figure 2-21 Small shim volume placed over the heart.



2. Place the parallel slices or single slice over the area of interest. Note that 2D gated FIESTA enhancement does not apply to radially prescribed slice groups.

IMPORTANT: To optimize the 2D gated FIESTA prescan enhancement, the intersection of the shim volume and slice locations should only be placed over the anatomy of interest.

Figure 2-22 Left = correct intersection and placement of shim volume and slice locations, Right = incorrect intersection and placement of shim volume.



Prescan

The enhanced 2D gated FIESTA prescan only applies to the intersection of the shim volume and the center slice of the graphic prescription. This ensures that the acquisition uses relevant frequencies centered on the area of interest. Frequencies outside the FOV that used to cause the spectrum degradation are thus eliminated with this technique.

Results

The image improvements are demonstrated in all 2D gated FIESTA scans that have been acquired as parallel slices within a single group or a group with a single slice, such as serially prescribed short axis images.

Figure 2-23 Note the reduced susceptibility artifact on image on the right.

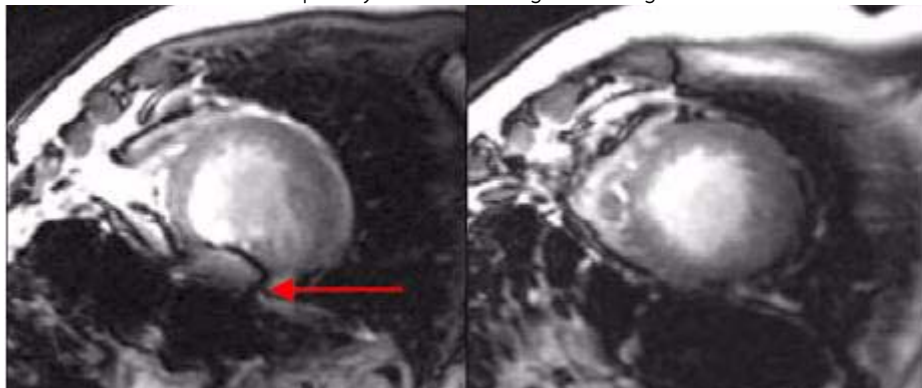
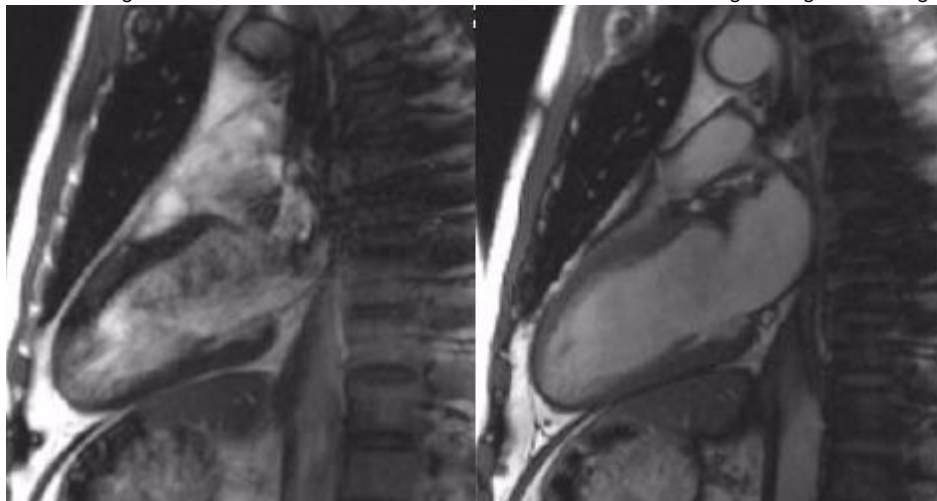


Figure 2-24 Image on left demonstrates off-resonance artifact on a 3.0T image, image on the right shows reduced artifact.



Applications

VIBRANT™ for Bilateral Breast Imaging

VIBRANT, the bilateral breast application, is only available with Invivo's 1.5T 4-channel and 7-channel breast coils, and with GEHC's 1.5T and 3.0T HD 8-channel VIBRANT Breast Array coils. These coils can be used to acquire either axial or sagittal bilateral or unilateral images on 1.5T systems. On 3.0T systems, VIBRANT is only available with sagittal imaging.

To shorten the scan time, axial VIBRANT automatically uses a partial Kz-filling technique that is a User CV for other 3D Fast **GRE** scans. Consider generating reformatted images from the axial high resolution delayed phase images.

Note that when using the 8-channel VIBRANT Breast Array coil for a sagittal VIBRANT acquisition, the scan may not complete reconstruction and the system may hang. This occurs if you exceed 48 slices in the calibration scan. **Therefore, do not prescribe more than 48 slices in the calibration scan.** Increase slice thickness to cover anatomy of interest.

VIBRANT WARNING

Zoom gradient mode is not recommended when acquiring a VIBRANT scan.



WARNING: Scanning VIBRANT with Zoom mode is NOT recommended. When zoom mode is used for breast imaging, there is a spatial distortion that can be seen on 3D data sets, especially in the lateral-most images. The distortion between 2D and 3D datasets can be up to 12 mm and is demonstrated in sagittal versus axial datasets. There is a potential risk for lesion localization misregistration during biopsy procedures which could result in having to re-biopsy the patient again.

Considerations

- ZIP x 2 is now available with axial VIBRANT acquisitions.
- When using the 8-channel Breast coil and a large number of slices per slab with multiple phases is prescribed, the reconstruction time can be very long. Therefore, with this dynamic dataset, it is highly recommended that ZIP x 2 be turned off.

The following scenario can cause a delay in acquisition due to data acquisition memory restrictions:

- A sagittal multi-phase VIBRANT scan followed by an axial series that has EDR turned on. Turn off EDR.
- When using ZIP x 2 with a large number of slices per slab on the contrast series, click **Prep Scan** and wait to confirm that the series has prepped before you inject contrast. If the series preps successfully, then you can be assured that the scan will complete uninterrupted.

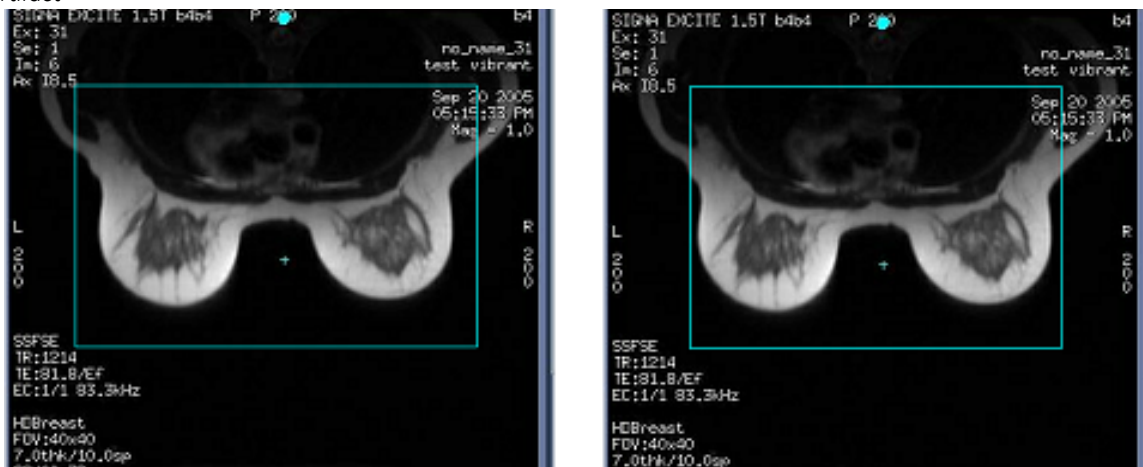
To minimize VIBRANT image artifact, keep the following in mind when prescribing the scan:

- Use a 48 cm FOV and slice thickness 15 mm for the ASSET calibration scan. For additional ASSET tips see [ASSET](#).

- For the sagittal series, use a wider right/left selection box to include the entire torso with extra slices at the end. For details see the following consideration.

An aliasing artifact from the slice direction can occur on VIBRANT sagittal breast images if the Graphic Rx 3D volume is placed too close to the edge of the anatomy. To avoid this artifact, increase the 3D volume.

Figure 2-25 3D volume Graphic Rx cursor position. Left = position to reduce possible aliasing artifacts, Right = position that may result in aliasing artifact



1.5T additional breast protocols.

1.5T Param	Sagittal Multi-phase	Axial Multi-phase	Sagittal T2	Ax T2	Axial Hi Res Delayed Phase
Application/PSD/Mode	VIBRANT/3D	VIBRANT/3D	FSE-XL/2D	FSE-XL/2D	VIBRANT/3D
Imaging Options	ASSET, NPW, Multi-phase	ASSET, Multi-Phase	NPW	ASSET, NPW	ASSET
TE & TR	Min Full	N/S	120-220/5000	120-220/6600	N/S
Flip Angle/ETL	FA: 10	FA: 10	ETL:20	ETL: 24	FA: 10
Bandwidth	31.25	41.67	31.25	31.25	62.50
Slices	92-108	60-80	92-108	60-80	60-80
Thickness	2.0-3.0	1.8-2.0	2.0-3.0	1.8-2.0	1.8
FOV	18-20	34-38	18-20	34-38	34-38
Frequency	256	384	256	320	512
Phase	160	320	224	224	350
NEX	1	N/S	2	1	N/S
PFOV	N/S	1	1	1	1
Freq Dir	A/P	A/P	A/P	A/P	A/P
Shim	Auto	Auto	Auto	Auto	Auto
Special	On	On	Off	Off	On
Turbo	2	2	2	2	
ASSET factor	2	2		2	2
Multi-Phase	Phases/Loc = 4-6	Phases/Loc = 4-6			

3.0T additional breast protocols

3.0T Param	Sagittal Multi-phase	Sagittal T2	Ax T2
Application/PSD/Mode	VIBRANT/3D	FSE-XL/2D	FSE-XL/2D
Imaging Options	ASSET, NPW, Multi-phase	NPW	ASSET
TE & TR	Minimum	120-220/5000	120-220/6600
Flip Angle/ETL	FA: 10	ETL:20	ETL: 24
Bandwidth	31.25	31.25	31.25
Slices	92-108	92-108	60-80
Thickness	2.0-3.0	2.0-3.0	1.8-2.0
FOV	18-20	18-20	34-38
Frequency	288	320	320
Phase	192	224	256
NEX	1	2	1
PFOV	1	1	1
Freq Dir	A/P	A/P	A/P
Shim	Auto	Auto	Auto
Special	On	Off	Off
Turbo	2	2	2
ASSET factor	2		2
Multi-Phase	Phases/Loc = 4-6 Variable Delay = On		

Breast Implant workflow

There are special factors to consider when scanning women with implants. Screening the patient for implant type is critical for selecting the Center Frequency peak in prescan and SAT pulse parameters. The prescan and SAT values determine the breast tissue and implant image contrast.

Screening questions for patients with breast implants

- How long have you had the implants?
- Are they silicone, saline, or both (double lumen)?
- Are both implants the same type?
- Do you think there has been a rupture?
- Has there been any change in the implants?

Center Frequency and SAT pulses

The following information can facilitate your understanding of setting CF in prescan and selecting SAT pulses during scan prescription.

Table 2-1 Frequency differences between silicone, fat, and water

Component	1.5T	3.0T
Fat	220 Hz less than water	440 Hz less than water
Silicone	approximately 320 Hz less than water	approximately 640 Hz less than water
Silicone from fat	100 Hz less than fat	200 Hz less than fat

Figure 2-26 Spectra illustrating the relationship of water (1), fat (2), and silicone (3) peaks

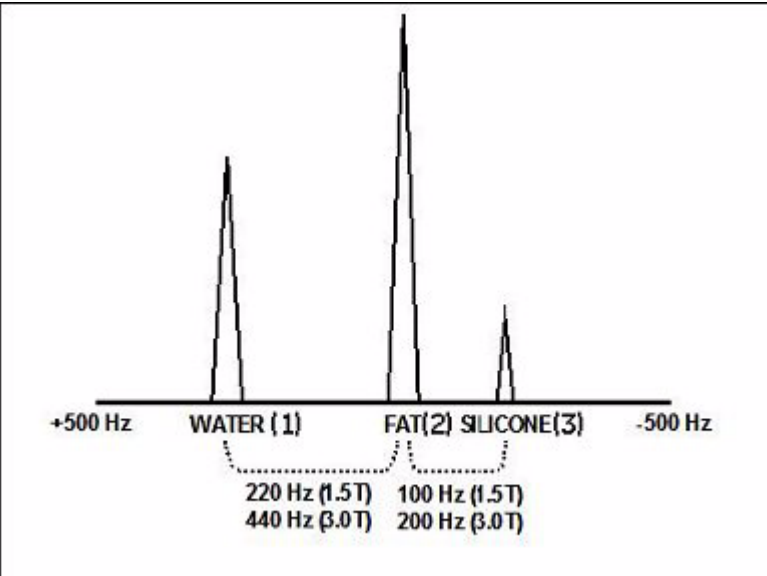


Table 2-2 displays the desired suppression technique and default center frequency upon completion of prescan.

Table 2-2 Suppression techniques at default Center Frequencies

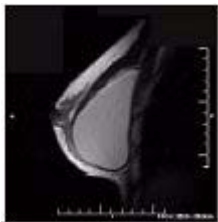
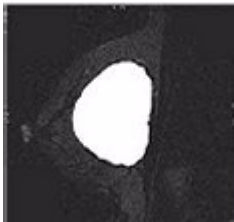
Suppression technique	Default CF	Frequency shift at 1.5T	Frequency shift at 3.0T
Fat saturation	Water peak	-220 Hz	-440 Hz
Water saturation	Fat peak	+220 Hz	+440 Hz
silicone saturation (choose fat saturation in the protocol)	Water peak	-320 Hz	-640 Hz

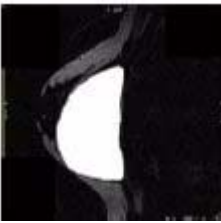
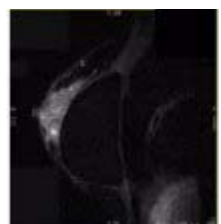
IMPORTANT:It is highly recommended that you perform manual prescan when imaging breast implants. Using the wrong center frequency can result in poor image quality.

Image contrast

Table 2-3 displays Silicone implant image contrast that is visualized with different PSDs. Saline implants have a different image contrast due to the implant's chemical composition.

Table 2-3 Silicone implant image contrast

Signal intensities	Fat	Water	Silicone
FSE T2 with no SAT pulse 	bright	bright	bright
FSE-IR with Water SAT pulse Applications include ruptures, dislocations, capsular contractures 	dark	dark	bright

Signal intensities	Fat	Water	Silicone
Silicone suppressed 	medium	bright	dark
FSE T2 with Water SAT pulse 	medium	dark	bright
FSE T2 with Fat SAT and suppression pulse centered over silicone peak Applications - imaging water filled with cysts 	dark	very bright	dark

Silicone implant procedure

MR imaging of silicone implants require high-resolution images. The silicone implant produces a low signal amplitude in the prescan. It has long T1 and T2 relaxation times that help to differentiate silicone from the fat and water peaks in prescan. The type of image contrast acquired is dependent on the clinical needs:

- fat suppression
- silicone suppression for evaluating fat or other tumors
- bright silicone contrast for evaluating prosthesis dislocation, intracapsular or extracapsular ruptures or capsular contractures

These protocols are specifically for silicone implants. In a breast with a saline implant, if you suppress water and fat, there is no signal from tissue. Therefore, it is only noise that creates some signal. All that is seen is skin contours and some ghosting artifacts from the heart.

Figure 2-27 Example of poor image quality when using a silicone protocol on a patient with saline implants



Bright silicone implant procedure

For optimum image quality, scan each breast individually in either the sagittal or axial plane.

When acquiring images that have a bright silicone appearance, the signal from both fat and water need to be suppressed. This can be achieved by following these steps:

1. Prescribe an **FSE-IR** (STIR) scan that nulls the fat signal and suppresses the water signal.
2. Select **Water** for the SAT pulse.
3. Complete Auto Prescan.
4. Select **Manual Prescan** and center on the fat peak.
 - The suppressed water peak is located at 220 Hz to the left of fat on 1.5T and 440 Hz to the left of fat on 3.0T.
 - The silicone peak is visible and it is located at -100 Hz to the right of the fat peak on 1.5T and -200 Hz to the right of the fat peak on 3.0T.
5. Click **Done** from Manual Prescan screen.
6. Acquire the scan.

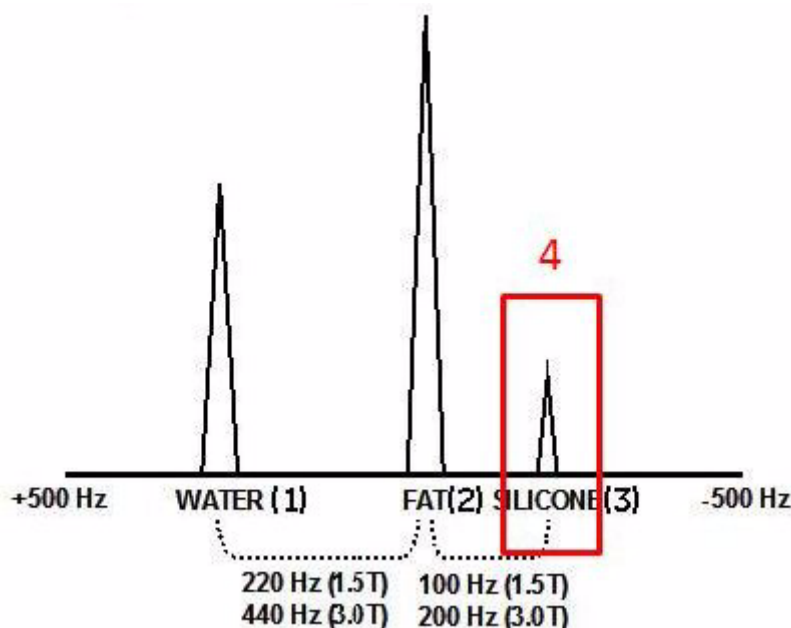
Method 1: Silicone suppressed implant procedure

Fat and silicone have similar resonance frequencies, therefore you must manipulate the center frequency so that you can successfully suppress the silicone peak.

1. Prescribe a **T2 FSE** sequence.
2. Select **Fat** for the SAT pulse.
3. Complete Auto Prescan.
4. Select **Manual Prescan**.
5. Select **Center Frequency Fine**.

6. Select the coil receiver that produces the highest signal.
7. Center the frequency over the **Water** peak. The default value for applying the fat suppression pulse is (-220) Hz for 1.5T and (-440) Hz for 3.0T. To suppress the signal from the silicone implant you need to adjust the location of the 100 Hz saturation pulse to be centered over silicone rather than fat.

Figure 2-28 Saturation pulse (4) centered over the silicone peak rather than the fat peak



8. Complete the following steps to adjust the saturation pulse over the silicone peak:
 - a) Click **Modify RSP** from the Options menu to view the Display RSPs window.
 - b) Type **csf** in the RSP Name type-in field.
 - c) Change the default value to center the saturation pulse over silicone rather than fat.
 - -320 for 1.5T
 - -640 for 3.0T
 - d) To see the saturation applied and the silicone peak drop, type **cstun** in the RSP Name field and type **1** in the Current Value field.
 - e) Click **Accept** to close the RSP screen.
9. Click **Done** to close Manual Prescan.
10. Acquire the scan.

Image appearance: silicone suppressed and signal from both fat and water. These images are useful for visualizing fat or lesions.

Method 2: Silicone suppressed implant procedure

Prescribe a STIR (FSE-IR) sequence with a TI = 400 ms.

Saline implant tips

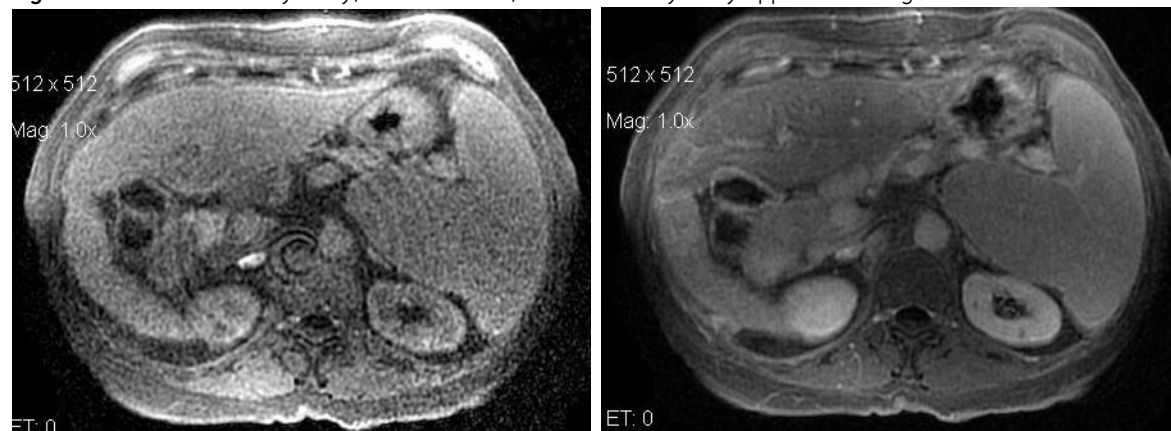
1. Prescribe a **T2-weighted FSE-XL** sequence.
2. Select one of the SAT pulses based on the clinical need:
 - Select **Fat** for saline implant only visualization and fat suppression.
 - Select **Water** for suppression of the saline implant to visualize breast tissue surrounding the implant.
3. Complete Auto Prescan.
4. Select **Manual Prescan** and center on the water peak.
5. Acquire the scan.

NOTE: For saline implant rupture visualization, contrast is not required.

LAVA

Frequently, when using the 8-channel Body Array coil, Full FOV with LAVA, a bull's eye artifact results. To avoid this issue, use upper or lower FOV for the 8-channel Body Array.

Figure 2-29 8-channel body array, full FOV on left, 8-channel body array upper FOV on right.



Imaging Options

No Phase Wrap

Combining NPW with an odd NEX or an even NEX that are not multiples of four NEX can result in a fine line artifact. Use one of the following methods to eliminate this problem:

- Use four, eight, or twelve (multiples of 4) NEX with NPW.
- Turn off NPW and use two, four, six, etc. (even) NEX, and either increase the FOV or swap phase/frequency direction to avoid phase wrapping.

Respiratory Gating and Triggering

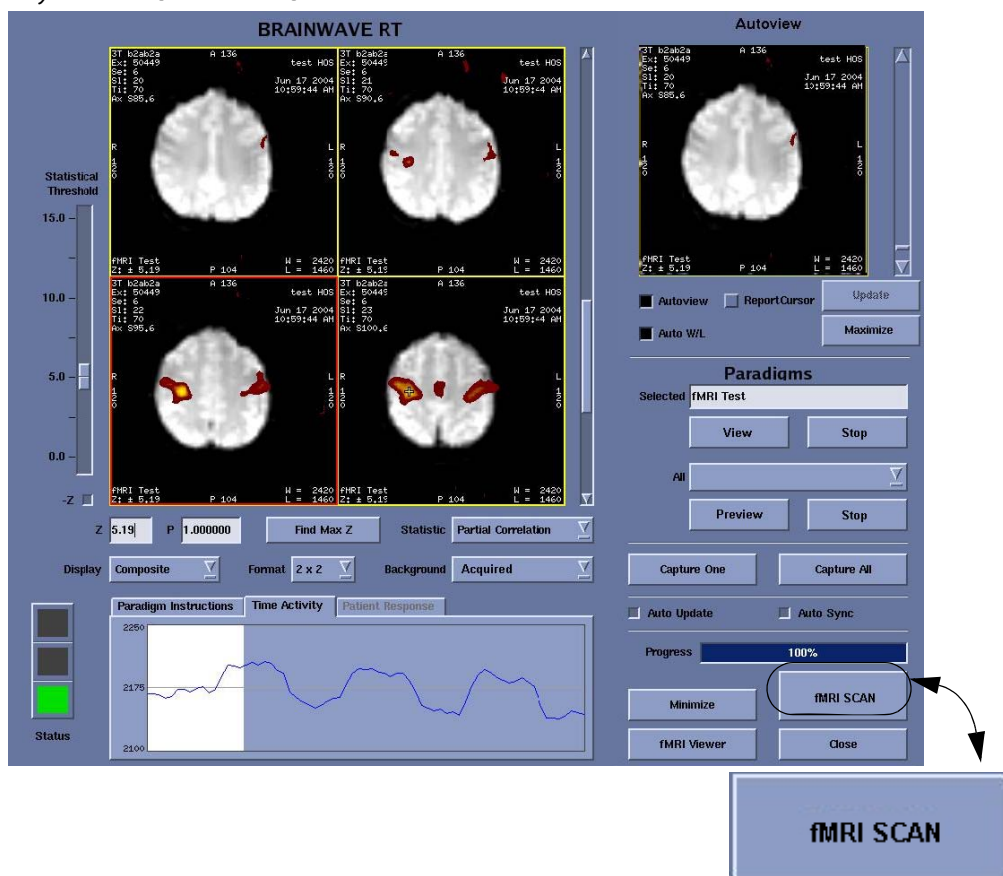
A trigger window of at least 80% should be used to acquire data during a small interval of respiratory cycle.

fMRI

If your system has the separate PC for paradigm development, during an fMRI acquisition, the Progress bar pauses and the paradigm window does not refresh properly. This only occurs on the operator console; the patient can see the paradigm very clearly through the goggles.

The paradigm window should be closed on the OC when BrainWave RT is playing the paradigm. When the paradigm window is closed, StimPC will talk only to BrainWave RT and the probability of this problem occurring is reduced. To see the paradigm play out, place the StimPC monitor next to the OC.

The Scan button on the keyboard is illuminated for an fMRI scan. When pressed, it will NOT start the fMRI scan. Only use the **[fMRI Scan]** button on the BrainWave RT window to activate the functional scan.



Do not use the following character in the Patient ID or Exam ID fields, if a BrainWaveRT series is going to be processed using BrainWavePA:

/, ', ", \$, &, |, \, *, ?, (,), {, }, [,]

Fluoro Trigger

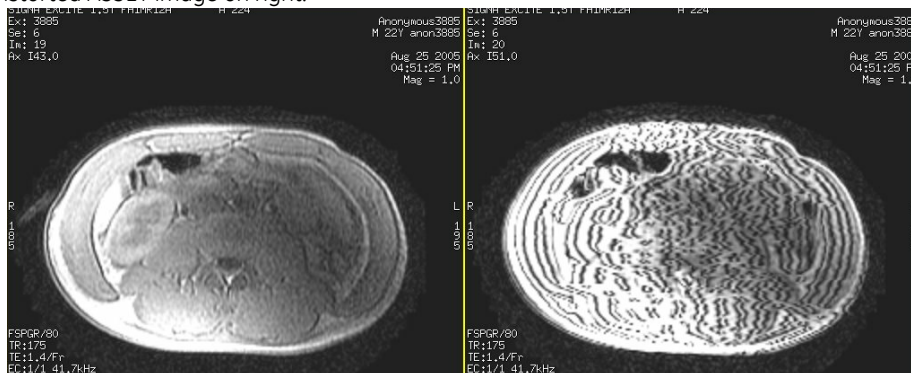
For the 3T fluoro trigger option, the default flip angle is too high when you initially enter the fluoro trigger screen. This flip angle results in poor differentiation between fresh flowing blood and arrival of contrast. Lower the flip angle on the fluoro trigger screen to distinguish between flowing blood and arrival of contrast.

ASSET

ASSET images may appear distorted when a too thin slice thickness is used in the cal scan. To avoid this issue, use a larger slice thickness for the calibration scan.

FOV	Recommended ASSET calibration slice thickness
30 cm	10 mm
48 cm	15 mm

Figure 2-30 Distorted ASSET image on right.



When the Imaging Options of ASSET and Respiratory Trigger are selected with a 2D abdominal scan, mis-registration occurs due to breathing motion. To avoid this issue, do not select the Respiratory Trigger Imaging Option with a 2D abdominal scan that has ASSET turned on.

Coils

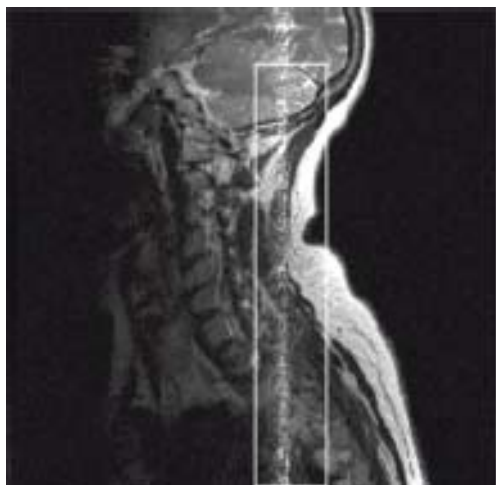
Annefact

Annefact artifact is a peripheral signal artifact that appears as ribbons of signal smeared through the image. This artifact is caused from signals generated outside the desired FOV that are detected by the receiver. Annefact appears in Fast Spin Echo (FSE) scans as smeared, bright, ghosting signals through the image in the phase direction. It typically appears on sagittal spines or pelvis scans using a phased array surface coil. Like a **Star artifact**, its origin is far from isocenter, where the gradients are nonlinear.

Uncompensated eddy currents in this area cause phase errors in the compressed signal and smear it through the image. Ways to prevent or reduce the incidence of the Annefact artifact include:

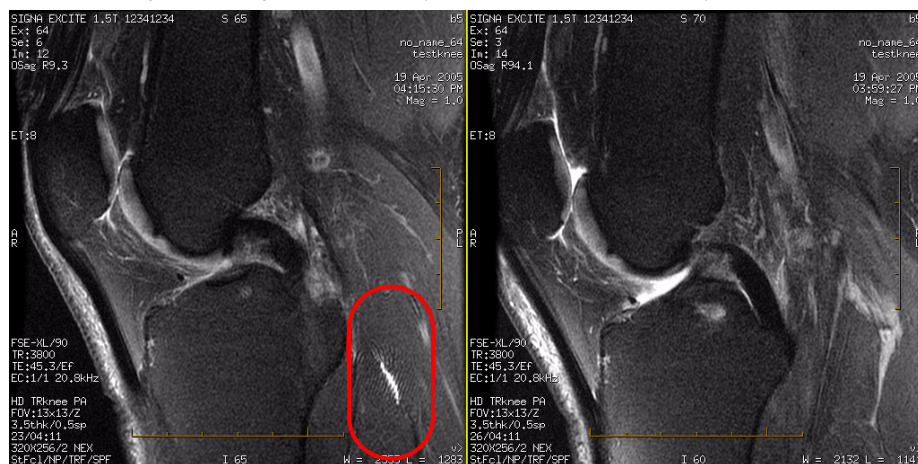
- Select the receive coils that match the imaging FOV (i.e., LS45, LS56, CS12, etc.)) which lessens the likelihood of picking up the peripheral signals that are generated outside the FOV
- Verify the frequency is set to S/I for sagittal spine imaging
- Place SAT band anterior to spine

This FSE sagittal cervical spine shows annefact. The image was acquired with CTLTOP, 24 cm FOV, and the phase and frequency swapped. The artifact could have been prevented by not swapping phase and frequency and using a 2-coil selection, such as CS12 so the surface coil coverage would closer match the scan FOV.



If the HD Knee Array coil is placed at isocenter, an annefact artifact can appear on the sagittal or coronal knee images, particularly with TwinSpeed systems. To avoid this issue, place the HD Knee Array coil right or left off-center by approximately 60-70mm (the positioning range is 60-120mm).

Figure 2-31 Note the annfact artifact on the left knee image that was acquired with the HD knee coil positioned at isocenter. The artifact is eliminated on the image on the right that was acquired with the HD knee coil positioned off-center.



When using the CTL coil combined with Zoom mode on a TwinSpeed system and phase has been swapped to S/I for a sagittal spine image, annfact artifact may occur. To avoid this issue, use Whole Body mode when swapping phase/frequency on a sagittal image acquired with the CTL coil. The limitations of the Zoom mode are described in the table below to help ensure the Zoom gradient is used when it could improve image quality over Whole Body mode.

Coils and Planes	Maximum FOV S/I	Maximum FOV A/P	Maximum FOV R/L
All coils except CTL All Planes	35	44	44
CTL Coil Sagittal and Sagittal Oblique Frequency=S/I (unswapped)	25	44	N/A
CTL Coil Frequency A/P = (Swapped)	Choose Whole Body	Choose Whole Body	N/A

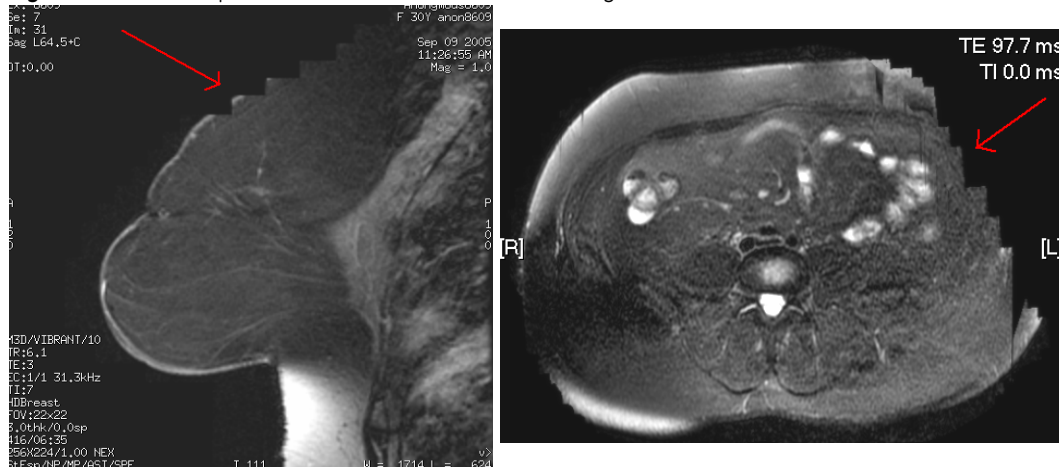
Coil artifact

When a stair-step type of artifact appears on the edge of the anatomy, it is due to one of the following:

- An element of the multiple-channel receive coil may be damaged or the connector for the coil is loose.
- The calibration extends too far beyond the coil.

Have your service engineer check your coil for bad channels. Do not acquire data beyond the calibration range. If you need to image this area, repeat the calibration scan and extend the volume to include the region of interest.

Figure 2-32 Stair-step artifact on breast and abdomen images.



Coil tips

Multi-channel coils such as the 8 channel CTL phased-array coil, have several different coil configurations. When moving from one coil region to another (e.g. from thoracic to lumbar regions), it is necessary to run autoprescan again (click **Auto Prescan > Scan**), since the coil configuration is different. If you choose to run manual prescan when moving to a new region (e.g. to the lumbar region from the thoracic region), you must select appropriate values of R1/R2. If not selected appropriately, an over-range will occur.

If you plug in an HDx coil into an HD system, the LED on the coil carriage turns green, indicating that the first level coil related check has passed. More checks are performed during scan prescription, which will result in the scan prescription failing since an HDx coil does not work on an HD system. Even though the light is green, scanning is prevented. A compatible coil must be plugged in and selected from the scan desktop in order to successfully start a scan.

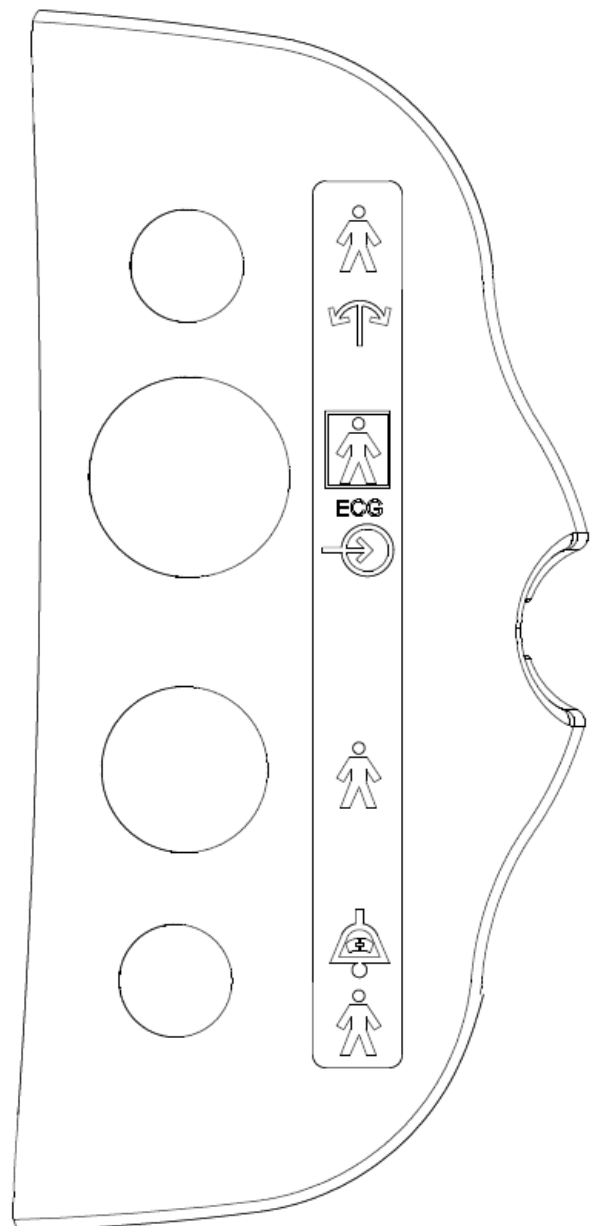
Equipment

PAC unit

A label has been added to the PAC unit to facilitate identifying each port.

Figure 2-33 PAC unit with label

Symbol	Port
	Respiratory Bellows Port
	ECG Leads Port
	Peripheral Gating Port
	Patient Alert Port



Respiratory Bellows

NOTE: The respiratory bellow and/or tubing may be seen on clinical images. Be sure to move the tubing if it is within the area of interest.

ECG lead wires

Only use the MRI High Impedance ECG Patient Lead Wire and Cable Set, orderable through the GE Accessories catalog, part number E8819RZ.

Dielectric pads

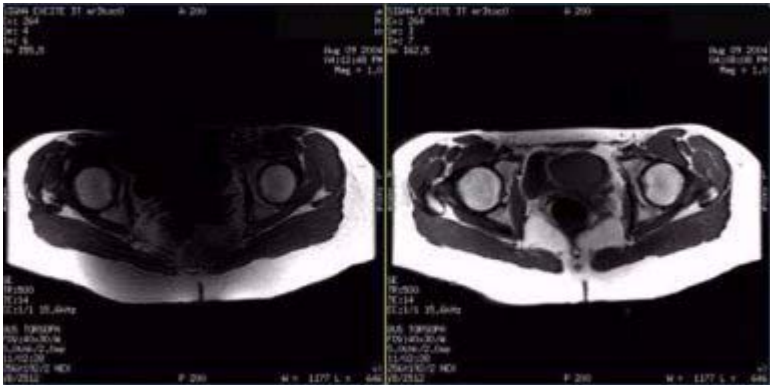
The Abdomen and Neck dielectric pads are used to suppress the effects of artifacts that can sometimes be encountered at higher field strengths. These effects can manifest as shading in the image, particularly when imaging in the cervical spine and pelvis. The amount of shading may depend on patient body habitus. It is usually worse with smaller patients, or in patients with ascites.

Applications

Dielectric pads, positioned between the coil and the patient, can minimize dielectric effects, such as shading seen in the image below.

- Use the Abdomen dielectric pad for abdominal and pelvis examinations.
- Use the Neck dielectric pad for cervical spine and neurovascular examinations.

Figure 2-34 Axial pelvis without dielectric pad (left) shows shading vs. pelvis with dielectric pad (right)



Specifications

Table 2-4 Dielectric pad specifications

Pad	Dimensions	Weight
Neck pad	4 inches × 16 inches (10.16 cm × 40.64 cm)	1.8 lbs (0.8 kg)
Abdomen pad	12 inches × 17 inches (30.48 cm × 43.18 cm)	5.5 lbs (0.25 kg)



CAUTION: Do not use these pads together with sharp objects. Refer to the Material Safety Data sheet for further information.

Replacement pads may be ordered through GE accessories:

- Complete set (includes neck and torso pads) - E 8823JB
- Torso pad - E 8823JA
- Neck pad - E 8823J

Dielectric abdomen pad procedure

Place the **Torso pad** lengthwise, between the patient and the coil, over the region of interest.

Figure 2-35 Torso dielectric pad placement



Dielectric neck pad procedure

Place the **Neck pad** lengthwise, between the patient and the coil, under the patient's neck.

Figure 2-36 Neck dielectric pad placement



Dielectric shading tips

Pockets of signal loss and non-uniformity can be observed on 3.0T images, which are attributed to the exacerbated dielectric resonance effect seen at higher field strengths. This effect is due to the interaction between the RF field and the patient. The wavelength at 3.0T is comparable to the patient body diameter, therefore it is more evident at 3.0T than at 1.5T. Dielectric shading is most troublesome for highly RF intensive sequences, small patients, or patients with ascites. It can also be intensified with multi-channel coils due to the increased signal making the shading more apparent.

Dielectric can be seen a various exams, including abdomen, pelvis, spines and runoffs. Dielectric effect can be reduced with the use of dielectric pads. Place the pads as directed.

Troubleshooting tips:

- Use non-conductive or dielectric pads, positioned between the coil and the patient, to minimize the effect.
- SCIC can be applied to compatible surface coil scans to correct for the low spatial frequency intensity modulations, reducing the dielectric effect.
- PURE can be applied to reduce coil intensity variations through a calibration process.

Figure 2-37 Axial hip acquired without a pad (left), with a pad (middle), and with a pad post SCIC (right)



When scanning runoffs on 3.0T, shading can occur primarily on the second station. This phenomenon manifests primarily as a right/left difference. There are a couple of ways to help reduce the amount of shading seen:

1. When setting up each station in the runoff, increase the overlap between the volumes (> 5 cm). For example:
 - Top station: place the superior aspect of the volume just above the renal arteries
 - Middle station: place the superior aspect of the volume at the bifurcation
 - Bottom station: place the superior aspect of the volume at or above the reference line
2. Place the dielectric pad LENGTHWISE over the patients groin.
3. Scan a 2D TOF gated of the middle station prior to scanning the 3D TOF
 - If shading is observed on the 2D TOF, be aware that the 3D TOF is likely to exhibit the same shading.

Figure 2-38 Run-off prescription

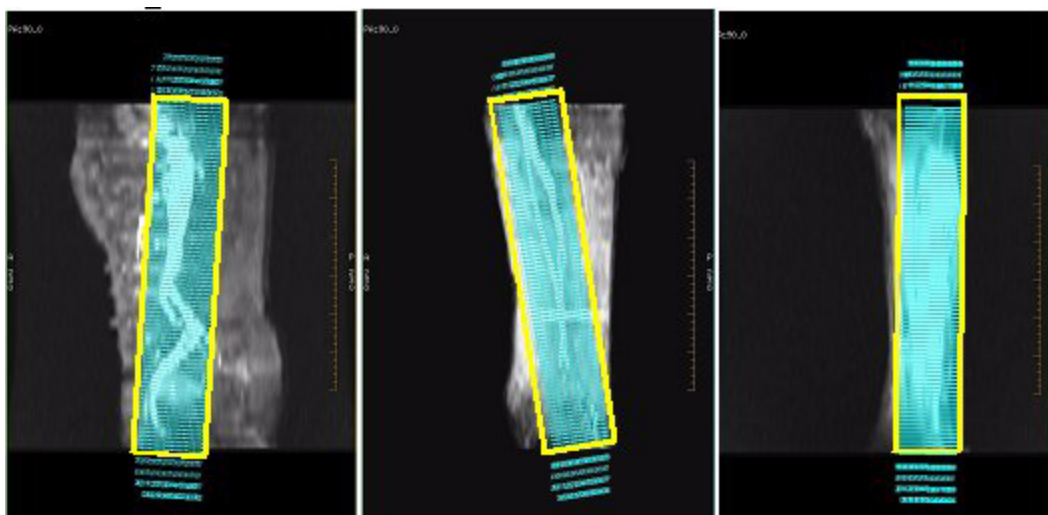


Figure 2-39 Example prescription of middle station volume placed on the top station localizer



User CVs

Flow Quantification Optimization

Flow Quantification Optimization (0=off, 1=on) is an acquisition parameter to reduce the phase error in quantitative flow measurement for the following PSD selections:

- Mode = **2D**, PSD Family = **Vascular**, PSD = **Fast 2D Phase Contrast**

Applications

Reduces phase error in flow measurement when used with Flow Analysis.

Considerations

When Flow Quantification Optimization is **On**, the TE is set to minimum and cannot be changed.

If the following conditions exist, the flow encoding gradients are optimized and the TE is lengthened by approximately 1 ms:

- Flow Recon Type = **Phase Diff**
- Flow Direction = **In Slice**
- Flow Analysis = **On**
- Flow Quantification Optimization = **1 (On)**

Classic Annefact Suppression

Classic Annefact suppression is a User CV only available on 3T systems for all 2D FSE-family pulse sequences (FSE, FRFSE, FSE-IR, T1Flair, and T2Flair) to suppress annefact/peripheral signal artifacts. You can turn the CV suppression technique on or off.

- Keeping the CV value at 1 (default) retains the Annefact suppression.
- Setting the CV to 0 turns off the suppression. Turning the suppression off can help alleviate the blurring/ghosting artifact that occurs at specific off-isocenter slice locations, which typically impacts one or two slices.

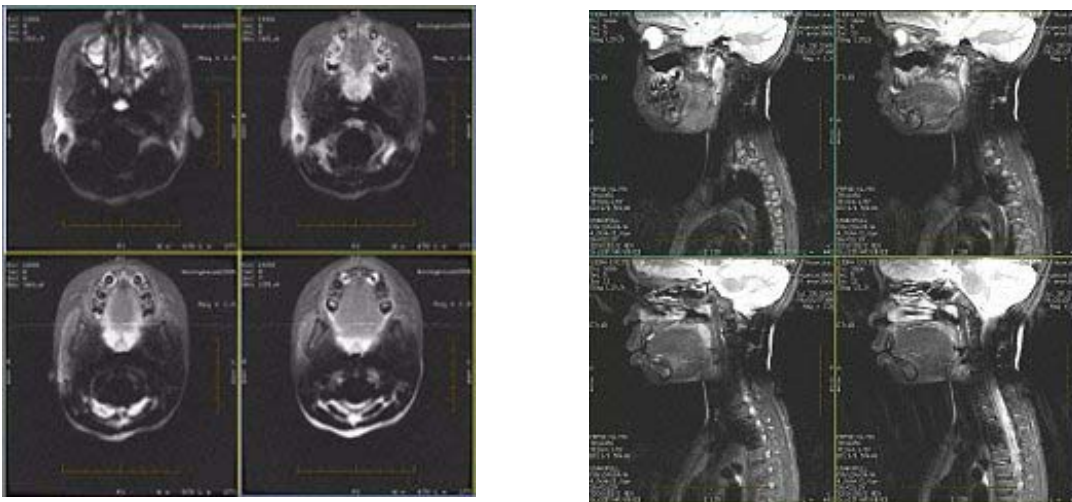
Bulk Susceptibility: fat SAT uniformity in C-spine

Poor chemical fat SAT uniformity can occur in C-spine imaging as a result of poor B_0 uniformity in the C-spine region. This is commonly referred to as a bulk susceptibility artifact.

B_0 uniformity in the C-spine is dependent on the patient's shape and size, as well as on the positioning of the patient within the magnet. A static magnetic field can become inhomogeneous within a patient due to irregular geometry and the distribution of material magnetic susceptibility (which is the physical property that determines how strong a magnetic field will be within a material). The areas of inhomogeneity result in poor fat suppression.

In anatomical regions like the C-spine, variations in the B_0 field exceeding 3 ppm can exist, which is approximately the fat-water separation range. Consequently, imaging sequences that employ chemical saturation to achieve fat suppression are extremely sensitive to B_0 inhomogeneity and can result in regions exhibiting partial or incomplete fat saturation.

Figure 2-40 Axial FSE (left) and sagittal FRFSE-XL (right) images displaying poor uniformity fat SAT



Troubleshooting Tips

- Use a STIR sequence, typically with a 140 to 160 ms TI time, instead of chemical fat SAT.

Chapter 3

Display

Introduction

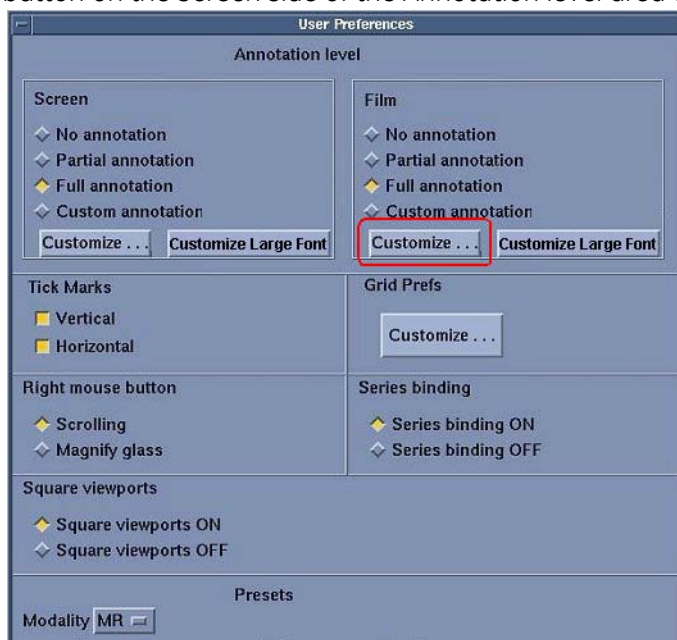
This chapter includes considerations for the following topics:

- [Filming considerations](#)
- [FuncTool considerations](#)
- [Display considerations](#)
- [Viewing a PDF document](#)

Filming considerations

When using the Multiple Image Display feature (2 on 1 or 6 on 1) for filming (F3 keyboard or MID Viewer button option), the system automatically adjusts the image annotation to a partial annotation level. Several image annotation fields are not displayed since the film real estate is limited. For example, scan time, matrix, NEX, coil, FOV, number of slices and imaging options are not displayed. These parameters are not displayed even if you have selected Full Annotation from the User Preferences screen. To view specific annotation complete the following steps.

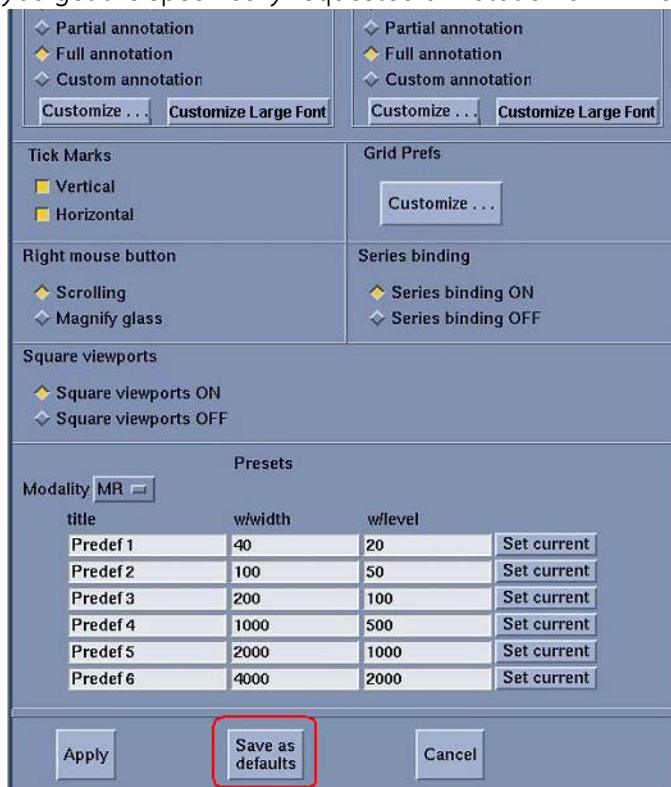
1. From the Browser, select an exam in which you want to use the MID film feature.
2. Click **Viewer** from the Browser list of applications.
3. From the Viewer control panel, click **User Preferences**.
4. From the User Preferences window, click **Customize** in the Film area of the Annotation level section.
 - Note that if you want to view the annotation levels on the screen, you must select Custom Annotation button on the Screen side of the Annotation level area of the User Preferences screen.



- From the Film/screen annotation groups menu, click all the desired annotation radio buttons that you want to appear on the film.



- Click **OK** from the Film/Screen annotations groups window.
- From the User Preferences window, click **Save as defaults** if you want this film annotation level to be the default setting. If not, click **Apply**, and the annotation level will only apply to the current application.
 - Note, do NOT select Full Annotation from the Annotation level area of the User Preferences window. This will give you a partial annotation level for the MID application. Only by using the Custom annotation level will you get the specifically requested annotation on film or screen.



FuncTool considerations

DTI

Non-axial EPI scans are subject to increased susceptibility artifacts that may not be correctable by the distortion correction.

ADC and eADC maps

CSF appears black in the ADC and eADC maps if the threshold is set improperly. The threshold can be improperly set with the auto launch of DWI in FuncTool. If this occurs, manually recalculate the maps in FuncTool.

Spectroscopy localizer

DWI images cannot be used as a spectroscopy localizer in FuncTool. Note that you can prescribe a spectroscopy scan from a DWI image, but FuncTool will not display it as the localizer. From the scan desktop, use a non-DWI image as the localizer for placing a spectroscopy VOI.

DTI Tracks

If you have turned off some of the tracks on a DTI FiberTrak image, and then right-clicked to Save the image, after the image is saved, all the tracks are automatically displayed again. In the upper right-hand corner of the fibertrak image, right-click to display a pull-down menu. Click **Tracts** to turn tracks back on or off.

DTI EPI distortion corrections

The correction function that occurs when you select **Apply Correction** from the Correction of EPI distortions FuncTool window, can automatically reduce distortions by scaling, de-skewing, and translating each image to align it with the reference image. The correction processing time can be as long as several minutes, depending on series size and processing speed. Non-axial EPI scans are subject to increased susceptibility artifacts that may not be correctable by the distortion correction.

Display considerations

When displaying images, consider the following:

- On occasion, the auto window level results in a bright then dark appearance of the images within a series. Manually adjust the window width and level.
- The image annotation for EPI images is FE (an abbreviation for Fractional Echo) for min TE and EF for min Full.

IVI

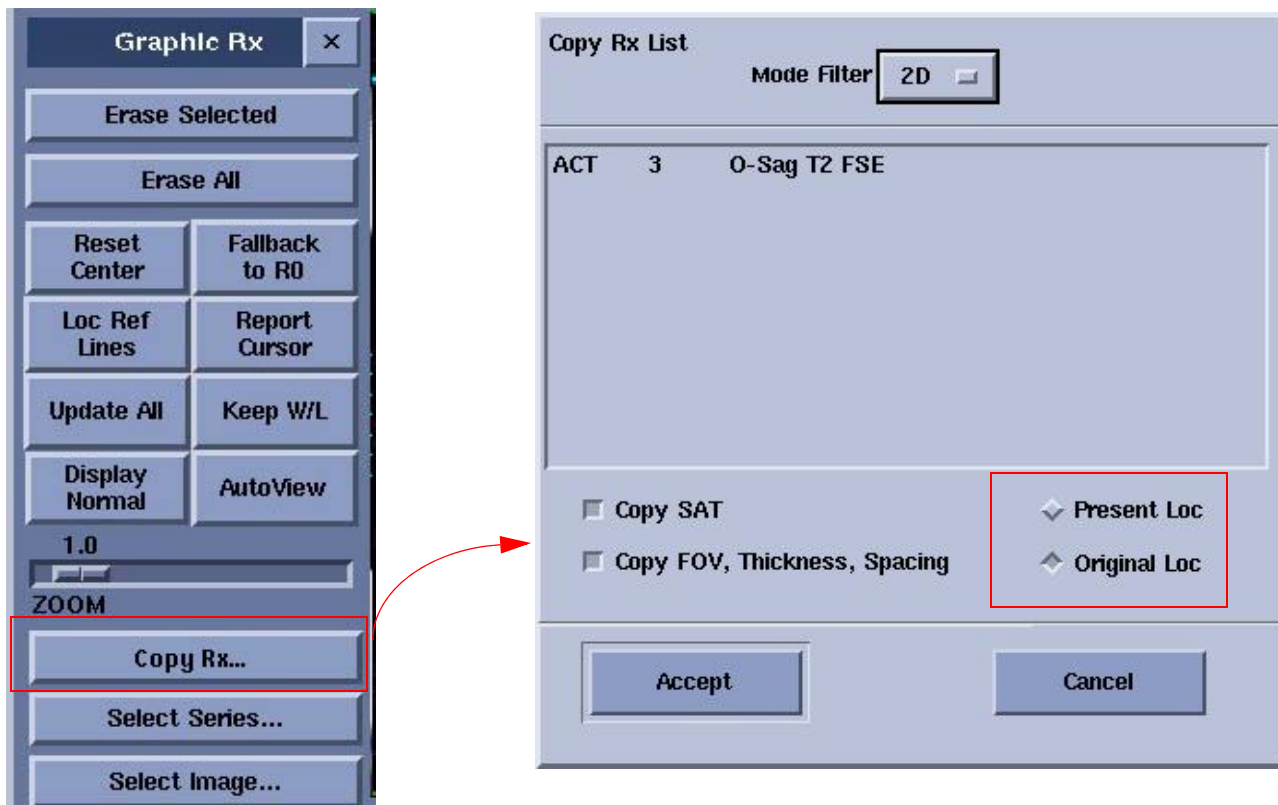
In IVI, there are Navigator protocols on the Additional Guides window. You can select these protocols and the Navigator wizard appears in the lower left viewport, but you cannot create a navigator view. Navigator application is not available on MR systems.



Add/Subtract

If you are using the Add/Subtract feature to subtract pre and post contrast series, follow this procedure to copy/paste the pre-contrast slice prescription lines to the post-contrast Graphic Rx localizer:

- From the Graphic Rx panel, click **Copy Rx**.
- From the Copy Rx panel, click **Original Loc**.
 - Selecting **Present Loc** or Reset Center will result in a mismatch location and will not allow subtraction of the pre and post contrast images.



Measurements

Image measurements are not possible on secondary capture images (e.g. screen saves). Secondary capture images do not contain the necessary spatial data to allow measurements. The add/subtract application generates secondary capture object types when performing the image combine operation, therefore, measurements will not be allowed on these images.

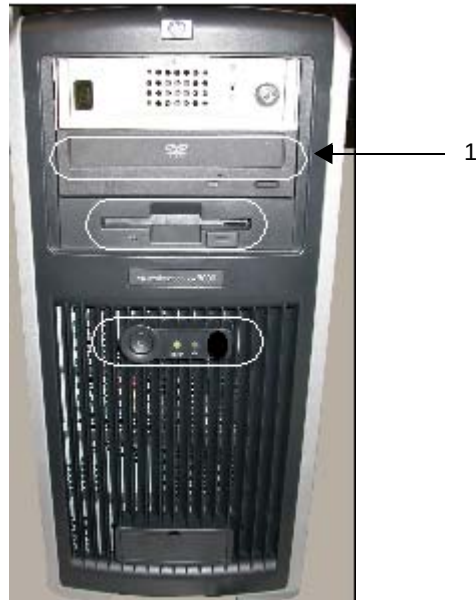
Viewing a PDF document

A PDF document that has been formatted for Adobe Acrobat® 4.0 can be viewed from the MR system.

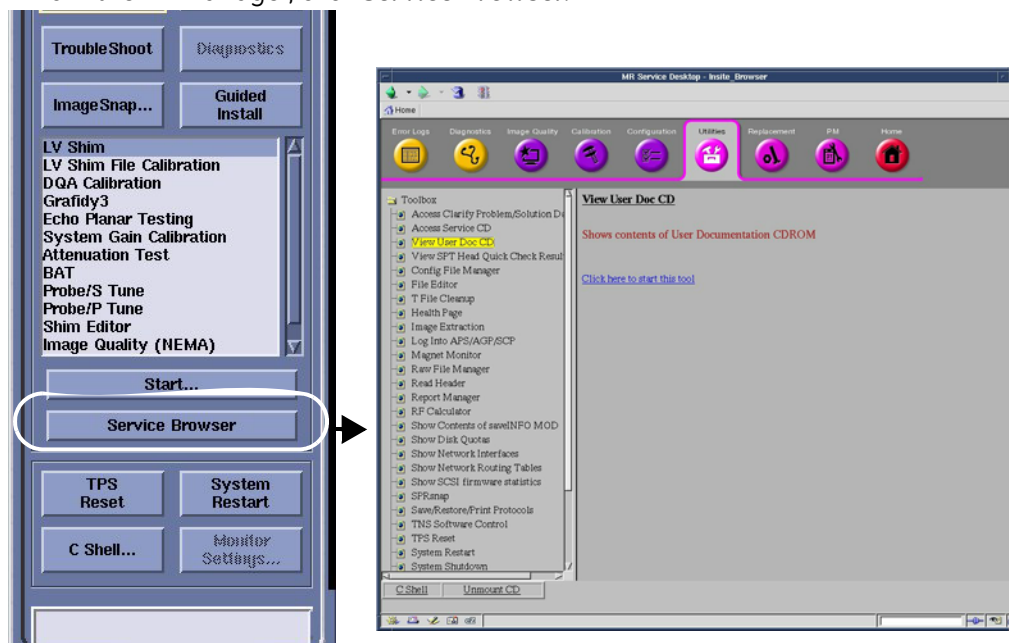
Start a PDF viewing session

1. Place the CD or DVD Adobe Acrobat 4.0 document, in the CD/DVD drive of the computer tower.

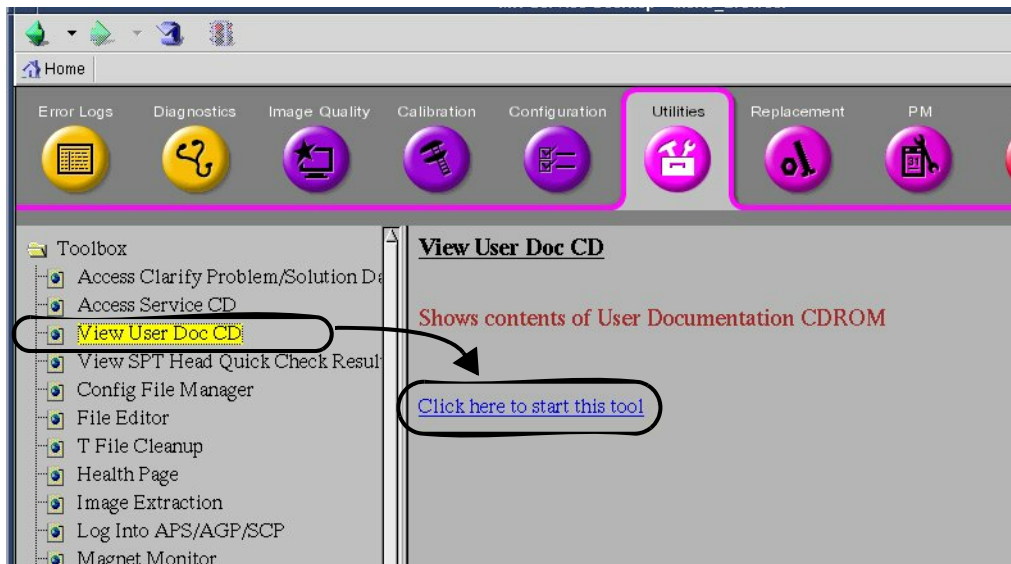
Figure 3-1 1 = CD/DVD drive of the computer tower.



2. From the Rx Manager, click **Service Browser**.



3. From the Service Browser click **Utilities** tab.
4. From the Service Browser Toolbox list, click **View User Doc CD** and then click **Click here to start this tool**.



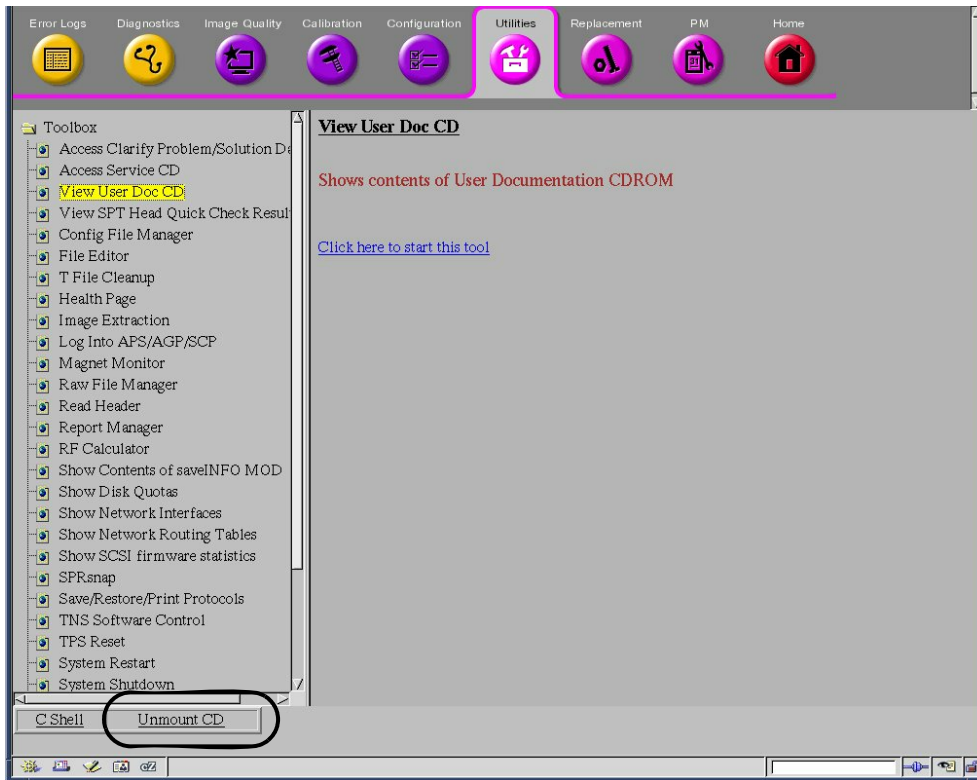
5. If this is the first time you have opened a User CD, then click **OK** to the Adobe Acrobat agreement.

End a PDF viewing session

1. Close the PDF document by clicking **File > Close**.
2. Close the Adobe Acrobat window by clicking **File > Exit**.

IMPORTANT: You must close the document and Adobe Acrobat before you proceed to the next step. If you do not close Adobe Acrobat, you cannot unmount the CD.

3. Click **Unmount CD** from the Utilities window of the service Browser.



4. Remove the CD/DVD from the drive.

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Chapter 4

Image Management

This chapter includes considerations for the following topics:

- [Structured Reports and GSPS considerations](#)
- [Auto Network considerations](#)
- [AW networked images](#)
- [Data Export](#)
- [Anonymize patient exam data](#)
- [CD Viewer](#)
- [Duplicate exam number](#)
- [Save Raw Data](#)

Structured Reports and GSPS considerations

- A DICOM Structured Report series generated from FuncTool or a Saved State series generated from the Viewer cannot be restored from an EXCITE HD MOD to a 9.1, 10.0, or 11.0 system. Nor can these series be networked from an EXCITE HD system to a 9.1, 10.0, or 11.0 system. When an EXCITE HD MOD is placed in a 9.1, 10.0, or 11.0 system, the MOD becomes a read only MOD. In other words, you cannot store 9.1, 10.0, or 11.0 images onto an EXCITE HD MOD.
- A 12.0 exam including SR or /GSPS object series to a 9.1 or 11.0 system is successful and no errors are reported. However, pushing or pulling a 12.0 SR or /GSPS object series from a 9.1/11.0 system will cause the exam to be paused on network queue with error to operator 'Receive failed.' Therefore do not network 12.0 GSPS or SR series to 9.1 or 11.0 systems.

Auto Network considerations

Auto network by series should transfer the images in the series once scanning has ended and the images are in the Browser. If an MR-Echo scan has been bookmarked and you immediately exit the series, the images may not transfer. If, by the time you have ended the exam and the series has not yet been networked to the end destination, manually transfer the series by completing the following steps:

1. From the Browser list, click the desired series to highlight it.
2. Click **Network**.
3. Click **Send Series**.

AW networked images

Image annotations unique to EXCITE HD such as VERSE, MART, LAVA, and MR-Echo when networked to a 10.0 or AW system do not display on the image. A software update is available for support of EXCITE HD annotation on EXCITE 11.0 and LX 9.1. Talk with your service engineer.

Data Export

Data Export current button doesn't update automatically when the following work flow is executed:

1. From the Compose tab, **Current** is selected as the Image Range Selection.
2. Save the current image and then page through the images.
3. Save another image.
 - The system does not automatically update to that image - instead, it resaves the previously saved image.

To avoid this issue, complete the following work flow:

1. Click **All** from Image Range Selection.
2. Reselect **Current** and it will update.

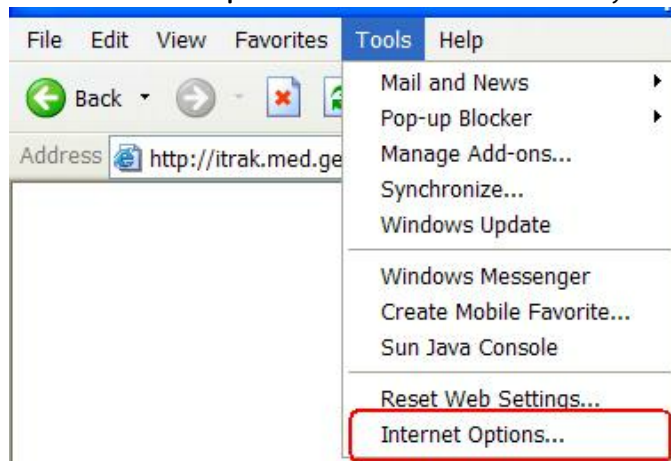
Anonymize patient exam data

Any series that is post processed (e.g., screen save, structured report, 3D images, GSPS series) cannot have anonymous applied. For example, if you anonymize an exam that has 5 series, one of which is a screen saved series, the anonymized exam in the Browser does not contain the screen saved series. The screen saved series has been deleted from the anonymized exam, but is still in the original exam.

CD Viewer

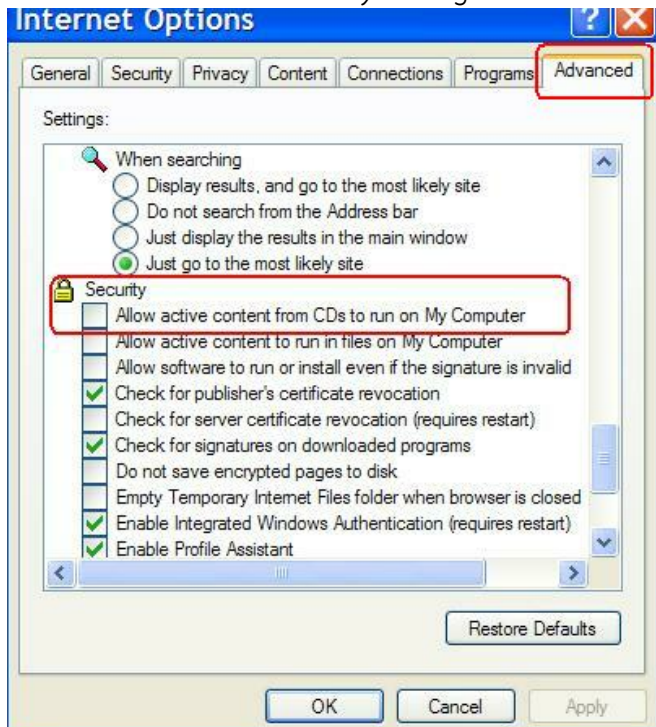
If you make a CD using the CD/DVD feature accessed from the Browser list of applications, and then load the CD onto a PC running Internet Explorer 7, an error occurs. The following steps allow you to display the CD Viewer on a PC running Internet Explorer 7.

1. Open Internet Explorer.
2. Select **Internet Options** from the Tools menu on your PC Internet Explorer menu bar.



3. From the Internet Options window, select the **Advanced** tab.

4. Scroll down the list to Security Settings and select **Allow active content from CDs to run on My Computer**.



5. Click **OK** to accept the new setting and close the Internet Options window.

Duplicate exam number

If you scan a patient twice by pulling the patient from the Modality Worklist, the result will be two separate exam numbers and subsequent series and images; for example, exam 1000 has series 1,2,3 and exam 1001 has series 1,2,3. When these images are networked to an AW or PACS, the two exams are combined and therefore the single exam will have two series one, two series two, etc.

To avoid this problem, when you need to rescan a patient due to some issue, do one of the following:

- Enter a new patient to the modality worklist so you do not need to reuse the one where the error (including system failure) occurred.
- Use Edit Patient Data on the first exam to regenerate a new study UID which will break the hard connection between the two studies.

Save Raw Data

To save raw data as DICOM images, perform these steps:

1. Open the Raw File Manager.
2. Select the Raw File that needs to be saved as a DICOM image from the list of P files displayed.
3. Click **Disk to Database** to save the selected P file to the image Browser.
4. Go to the image Browser and select the newly created series in the appropriate exam.
5. Open the CD/DVD tool.
6. Follow the instructions of the CD/DVD tool to archive a DICOM exam/series.

Chapter 5

fMRI Visual Paradigm Development

Introduction

Paradigms on the GE BrainWaveHW Lite stimulus PC are created and edited using the NeuroActivator software resident on the stimulus PC. Paradigms are then managed using the Paradigm Manager, which connects the paradigms on the GE BrainWaveHW Lite stimulus PC to the BrainWaveRT scanning acquisition, and allows them to be synchronized as hardware paradigms. If you are not using the GE BrainWaveHW Lite paradigm system, the Paradigm Manager will allow you to create and customize the timing for your software paradigms.

This document is a procedure for how to create a visual paradigm. For complete instructions about BrainWave, read BrainWave Learning and Reference Guide, direction # 5123077, Rev. 1.

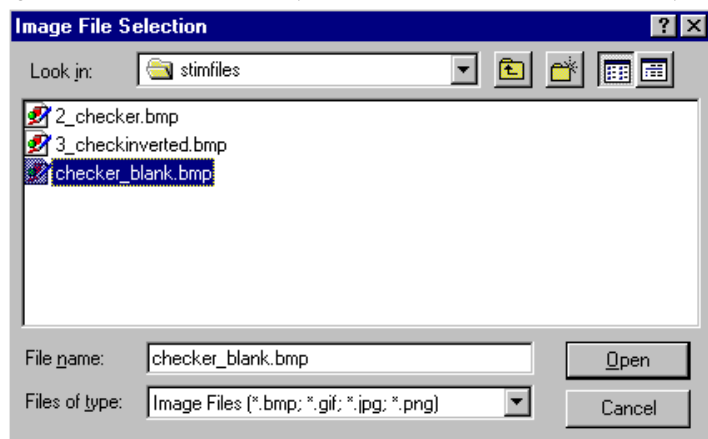
The Paradigm Manager allows you to create the following:

- Hardware Paradigms
- Software Paradigms
- Settings and Help

Generate a New Hardware Paradigm

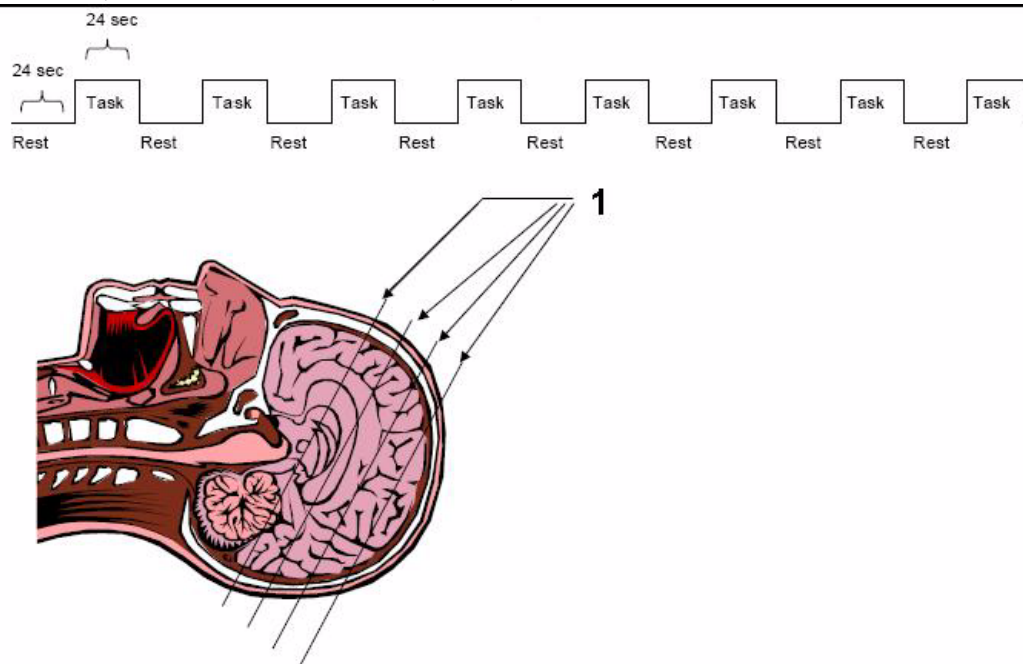
With BrainWaveHW Lite, you can create a new Hardware paradigm by making modifications to an existing one. This document will use an example of modifying a motor paradigm to create a visual paradigm. This sample visual paradigm uses three images.

Figure 5-1 Checker_blank.bmp for the rest state and 2_checker.bmp and 3_checkerinverted.bmp for the task state.



The paradigm is created with a 24-second task and rest time with a scan TR time of 4000 ms. There are eight rest and eight task cycles.

Figure 5-2 Rest/task cycle. If the TR = 4000ms and the task and rest periods is 24 seconds, there will be 6 phases for each rest and task period ($24/4 = 6$). 1 = Each of the 4 locations will have 6 phases per rest and task period. If there are 16 rest/task cycles and 6 phases per individual cycle, then there are a total of 96 phases per location.

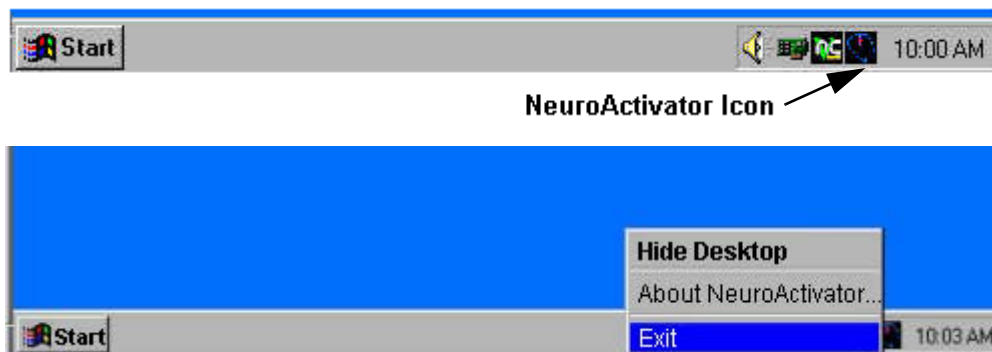
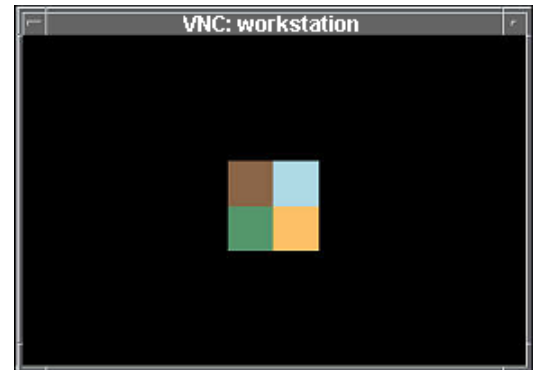


Prerequisite

Create a folder on the hardware PC desktop and transfer the images you want to use in the visual paradigm from a CD to the folder. Note that the images must be saved as a .bmp or .gif file type and not as a .jpeg file type. Check the image properties (right-click the file and select Properties) and make sure that it is not Read Only. If Read Only is checked, click the check box to deselect it.

Start StimPC

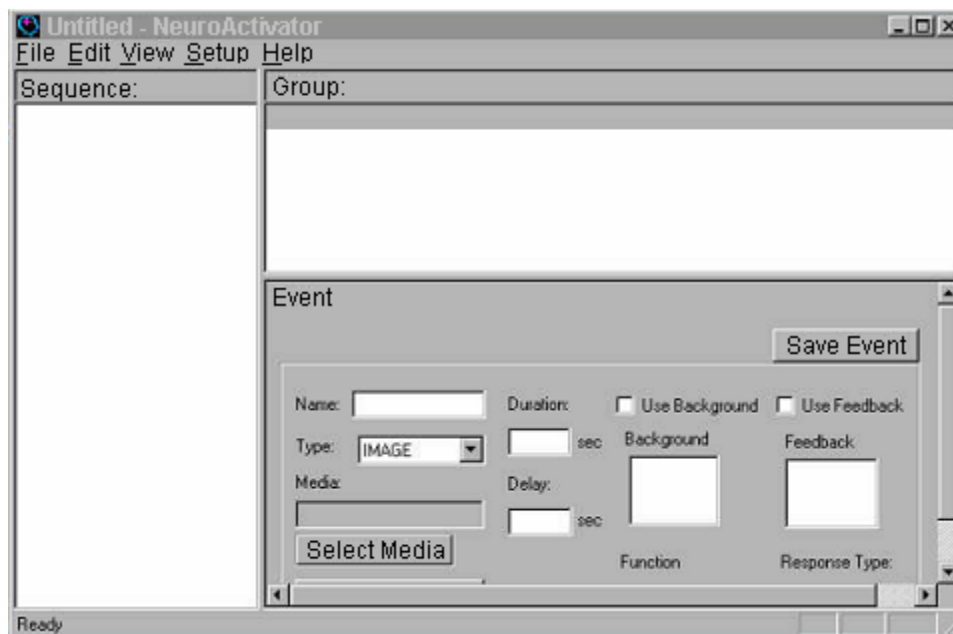
- You can complete the paradigm generation steps from the PC or from the MR console. If you are creating the paradigm from the MR console, click **[Open Console to StimPC]** on the Hardware tab of the Paradigm Manager.
 - The four box display of the NeuroActivator software appears.
 - If the **Open Window to StimPC** button does not open a connection to the Stimulus PC, perform one of the following suggestions:
 - Check the IP address on the settings page.
 - Open a command window and enter the command: **ping <ip address>** typically 10.0.1.9.
 - Make sure the Stimulus PC is operating. Confirm that only one version of the software is installed on the StimPC. On the StimPC, open a command window and type: `cd C:\Program Files\Sensor Systems, Inc.\NeuroActivator 1.050 w. Archiver 1.01` should be the only listing.
 - If you move the cursor into the PC window, the four box display is momentarily suspended.
- Terminate the Clinical mode of NeuroActivator by right-clicking the NeuroActivator icon in the lower right corner of the screen and selecting **Exit**.
 - The NeuroActivator icon is a small outline of a head on a black background.



3. Click **Start > Programs > NeuroActivator > NeuroActivator - Research**.

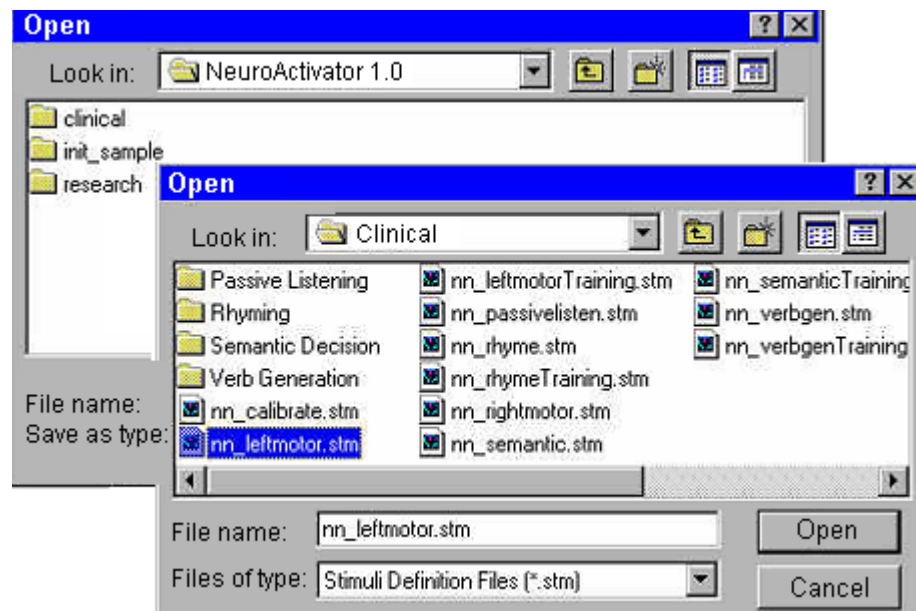


- If a License menu appears, contact your GE Service Engineer to get the license key installed.
- A window opens with three areas, titled Sequence, Group, and Event.

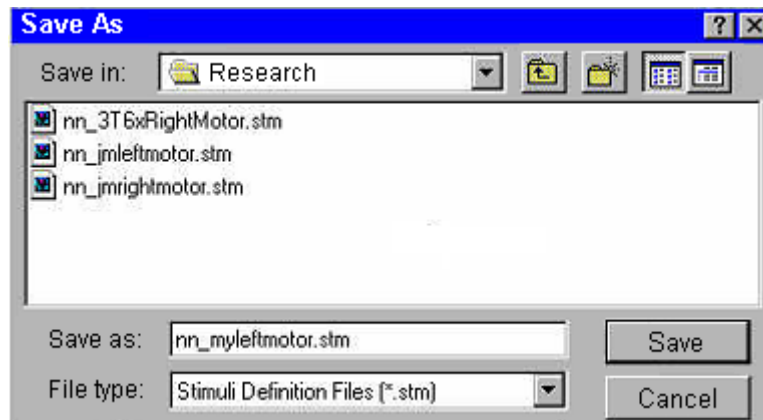


4. Select **File > Open** on the menu bar.

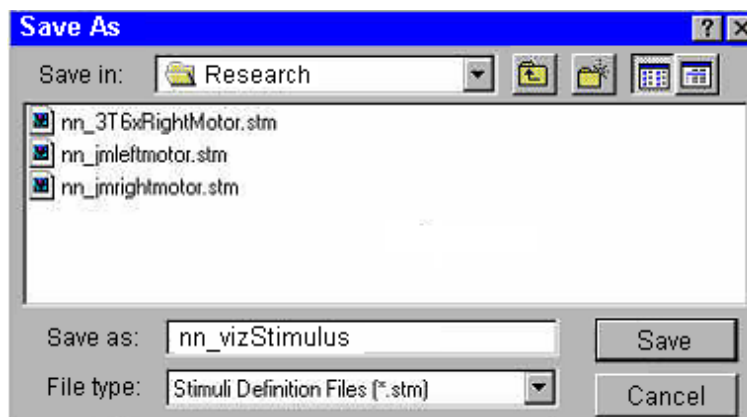
5. Navigate to the **Clinical** folder and select one of the existing paradigms.



6. Click **Open**.
 - The Sequence area displays the tree diagram for the selected paradigm.
7. Select **File > Save As**.
 - The Save As window opens.
8. Navigate to the **Research** folder.
 - All user-created paradigm (.stm) files must be stored in the Research folder.



9. Enter a unique file name.



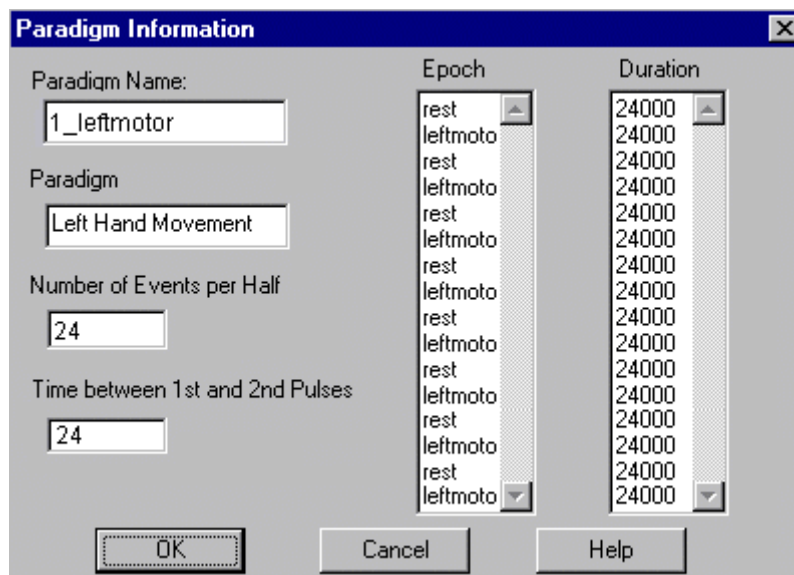
- The file name must be in the form of nn_filename.stm.
- The first letter of the paradigm name must be a number or lower case, e.g., 2134_myleftmotor.stm.
- If a number is used in the name it must be between 2000 and 2999.
- The file name must not have a period in the name, e.g., nn_viz.Stimulus is incorrect and nn__vizStimulus is correct.



WARNING: When creating paradigms, if you do not create a unique paradigm number, paradigm string, and paradigm name/filename, the Clinical mode of NeuroActivator may fail to communicate with BrainWave software. After editing paradigms, always be sure BrainWave communication with the clinical software is intact by opening the Paradigm Manager from the tools desktop and clicking [Refresh Hardware Paradigm List].

10. Click **Save**.

- The Paradigm Information window appears.



11. Enter the information appropriate to the paradigm you wish to create.

- Note that in the Epoch description in the above example, rest and leftmotor do not change automatically when you create a new paradigm. Therefore, if you want to change the Epoch name, double-click the item in the list and type the desired name. Next, copy the item and paste it over the desired item in the list. For example, if you want leftmotor to read task, then double-click the first leftmotor, type task, select task and right-click to copy it. Select each successive leftmotor and right-click to paste task.

Figure 5-3 Double-click the first name you want to change in the Epoch list. Copy this item and then double click each item you want to paste over the new name.

Epoch	Duration
rest	24000
task	24000
rest	24000
leftmoto	24000
rest	24000
leftmoto	24000
rest	24000
leftmoto	24000
rest	24000
leftmoto	24000
rest	24000
leftmoto	24000
rest	24000
leftmoto	24000

12. Click **OK** to accept the changes.

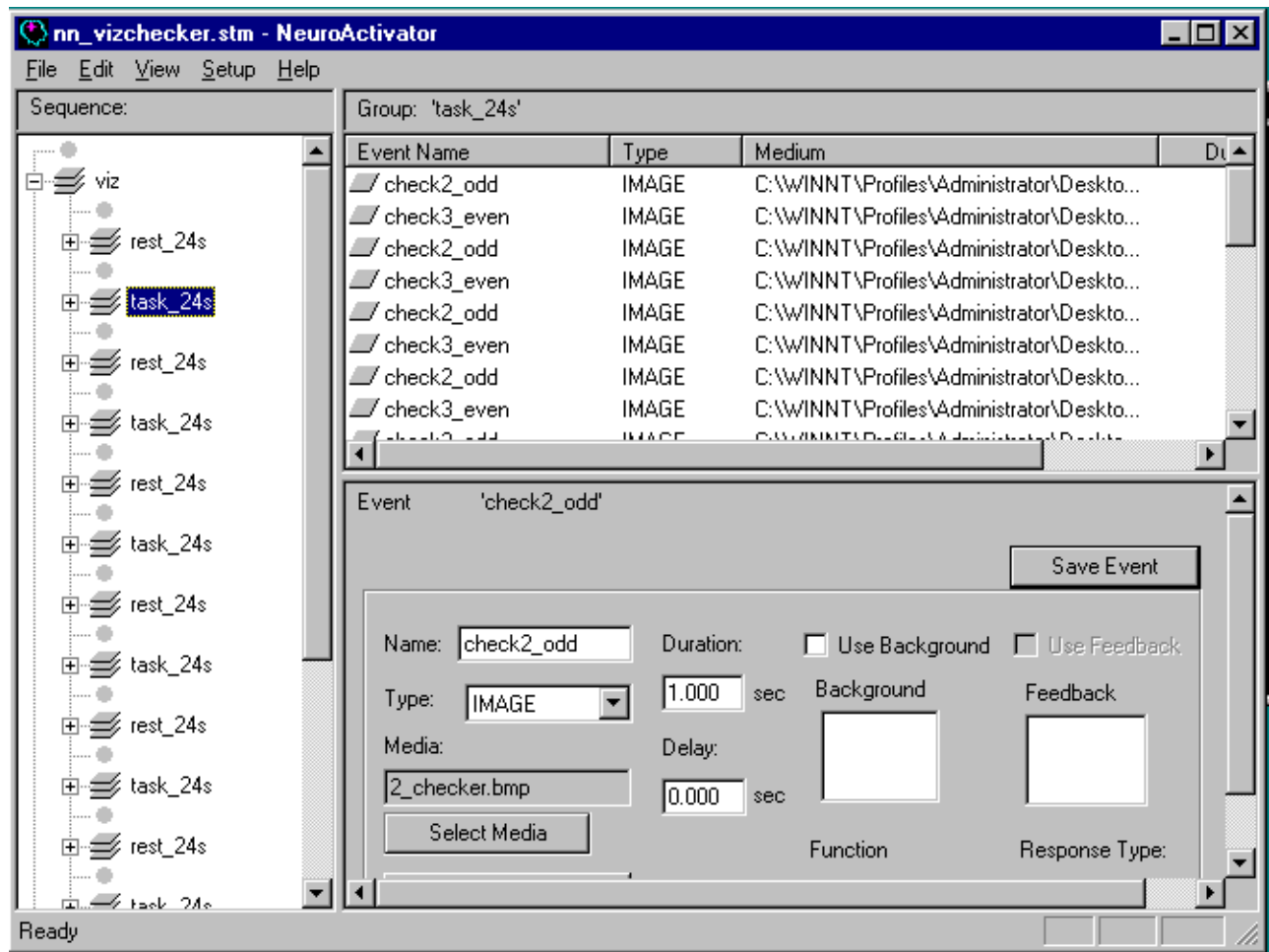
- The Paradigm Information window closes and you return to the main window.

Change the Paradigm Values

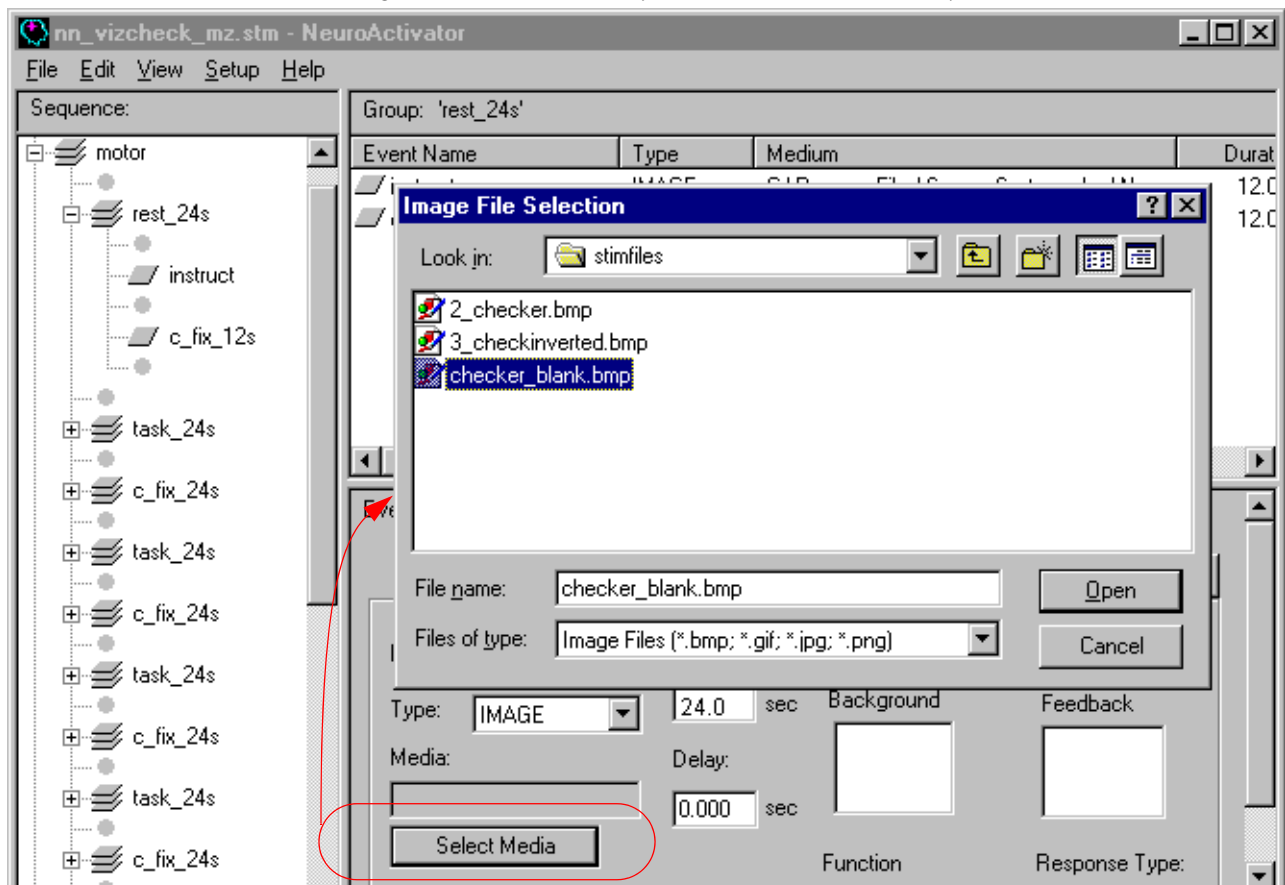
Follow these steps to change the paradigm values for each task in the sequence tree. The end result should look like Figure 5-4.

IMPORTANT:Change the values in the order listed and then Save Change after each task has been edited. There is no visual indication if you do not Save Change after modifying each task, but, if you do not Save Change, that task will not be included in the paradigm.

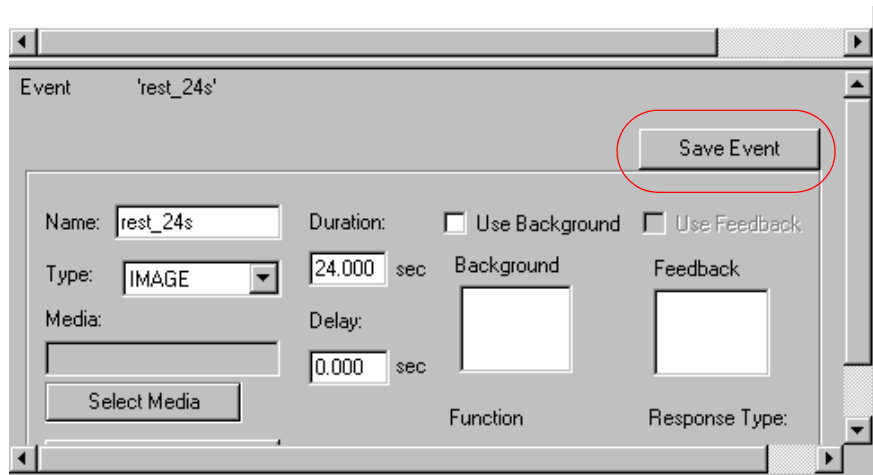
Figure 5-4 Visualization stimulus paradigm.



1. Click the plus (+) sign to open the Sequence tree on the left side of the Paradigm information window.
2. Place the cursor in the file text, right-click, select **Rename**, and type a new name. For example, change c_fix_24s to rest_24s.
3. Complete all fields on the lower half of the window.
 - Enter 24.0 for the Duration value for the rest event.
 - Enter 0.0 for the delay time.
 - Deselect Use Background check box. During the rest state, the patient will see the image entered from the Select Media file.
 - Click **Select Media** and navigate to the file folder you stored on the desktop. Select the desired rest image.

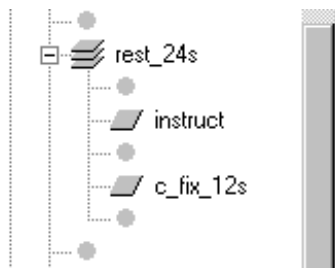


4. Click **Save Event**.



- Failing to save the event means that the event will not be part of the paradigm. It will appear in the Sequence tree, but it will not be applied to the paradigm during the scan.

5. Delete the other events in the Sequence tree under rest; in this example, delete instruct and c_fix_12s.



6. Select the next item in the Sequence tree; in this example c_fix_24s. Right-click and rename it task.

- The goal of the next step is to create two events in the task: one event with an odd-checkered pattern displayed for one second and one event with an even-checkered pattern displayed for one second. Once these two events have been created, you can copy, paste, **Save Event**, copy, paste, **Save Event**, etc. until you have twelve odd and twelve even events.

7. Select the first event in the task and change the name to check2_odd.

8. Complete all fields on the lower half of the window.

- Enter 1.0 for the Duration value for the rest event.
- Enter 0.0 for the delay time.
- Deselect Use Background check box. During the task state, the patient will see the image entered from the Select Media file.
- Click **Select Media** and navigate to the file folder you stored on the desktop. Select the desired task image; in this example, 2_checker.bmp.

9. Click **Save Event**.

10. Select the second event in the task and change the name to check3_even.

11. Complete all fields on the lower half of the window.

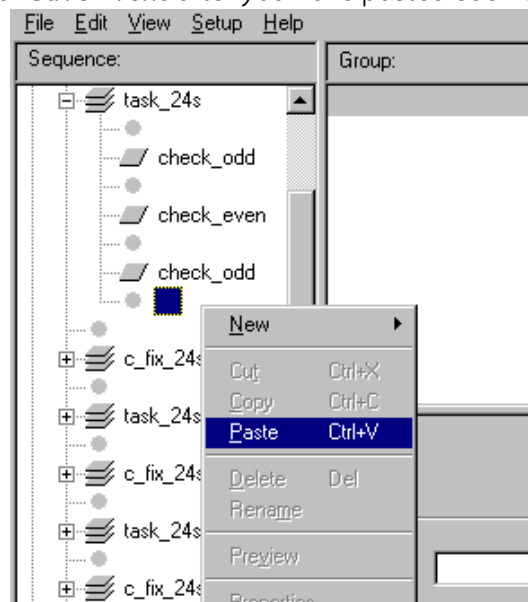
- Enter 1.0 for the Duration value for the rest event.

- Enter 0.0 for the delay time.
- Deselect **Use Background** check box. During the task state the patient will see the image entered from the Select Media file.
- Click **Select Media** and navigate to the file folder you stored on the desktop. Select the desired task image; in this example, 3_checkinverted.bmp.

12. Click **Save Event**.

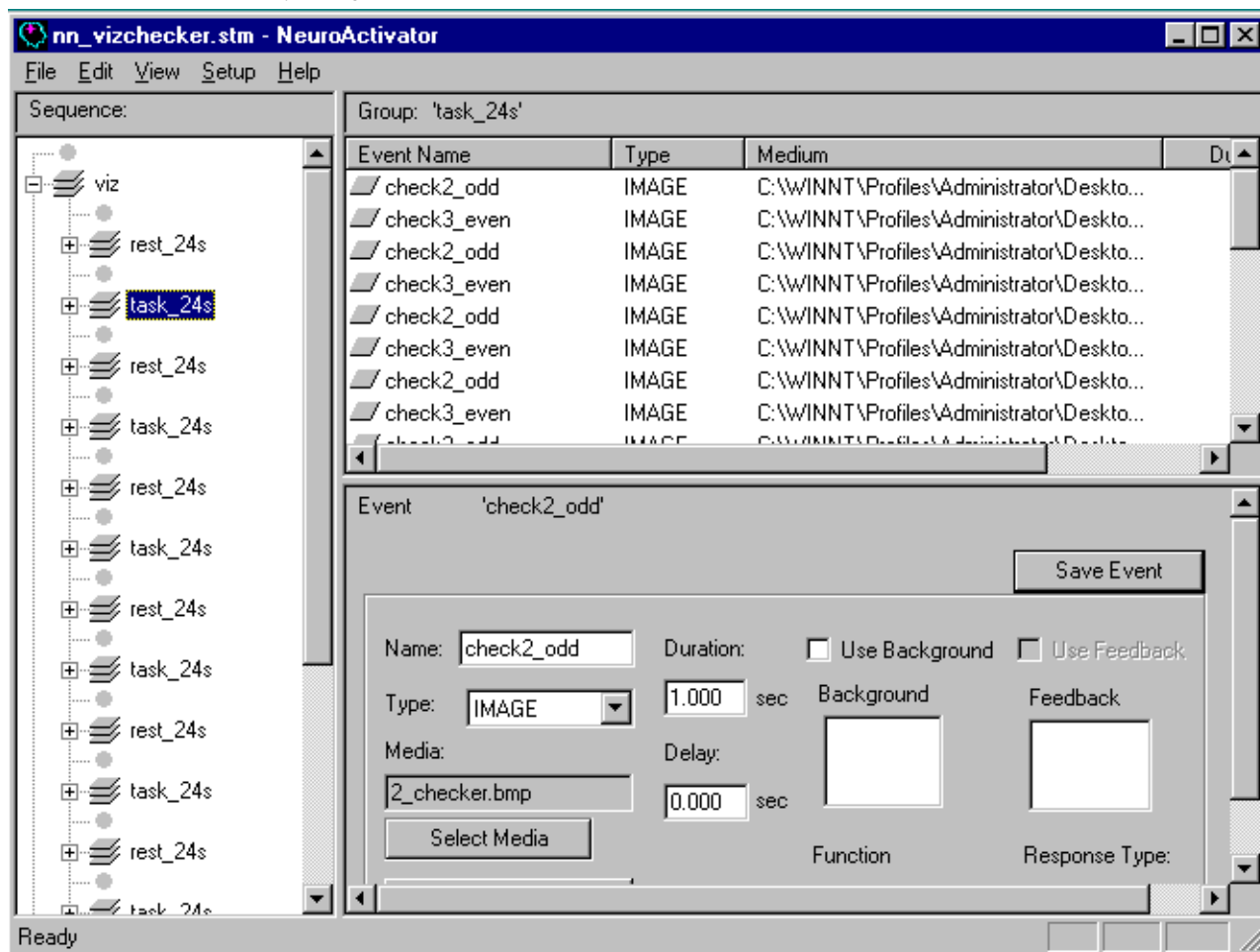
13. You now have an even and odd event in the task or stimulus epoch portion of the paradigm. To create 12 of each in the task, it is easiest to copy/paste an odd event and then copy/paste an even event.

- Be sure to click **Save Event** after you have pasted each event.



- Create 12 odd and 12 even events that occur for one second each so that there will be 24 seconds of checker pattern stimulus during the task epoch.
14. Delete all remaining items in the Sequence list except the End item. Select the item and press **[Delete]** on the keyboard.
- When the above task is completed, you should only have one visual rest epoch, one visual task epoch, and an end epoch in your Sequence tree.
15. Select the rest epoch, right-click > Copy, position the cursor after the last task epoch, and right-click > Paste.
16. Select the task epoch, right-click > Copy, position the cursor after the last rest epoch, and right-click > Paste.
17. Repeat the previous two steps until you have eight rest, eight task, and one end epoch.

Figure 5-5 Visualization stimulus paradigm.



18. Select **File > Save** on the menu bar.

- The Paradigm Information window appears again.

19. Make changes so the **Epoch** and **Duration** columns correspond with the previous sequence/group/event changes.

- In this example, the leftmotor epoch needs to be changed to task. Select leftmotor, right-click and type task.

Paradigm Name:	Epoch	Duration
vizStim	rest	24000
	task	24000
	rest	24000
	leftmoto	24000
	rest	24000
	leftmoto	24000
	rest	24000
	leftmoto	24000
	rest	24000
	leftmoto	24000
	rest	24000
	leftmoto	24000
	rest	24000
	leftmoto	24000
	rest	24000
	leftmoto	24000

20. Click **OK**.
 - The Paradigm Information window closes.
21. Select **File > Exit** on the menu bar.
 - The NeuroActivator window closes.
22. Select **Start > Programs > NeuroActivator > NeuroActivator - Clinical**.
 - The four box display of the NeuroActivator software appears.
23. Click **Refresh Hardware Paradigm List** on the Hardware Paradigms tab of the Paradigm Manager.
 - The paradigm names update in the Paradigm Names pull-down list to include your newly created paradigm.

- If the **Refresh Hardware Paradigm List** does not respond or generates an error, perform one of the following suggestions:
 - Check the IP address and communication port of the Settings and Help tab.
 - Open a command window and enter the command: **ping <ip address> typically 10.0.1.9.**
 - Make sure the Stimulus PC is on and operating the Stimulus software in Clinical mode.
 - Make sure the Stimulus PC software is configured with scanner host-name and IP address.
 - Make sure all paradigms have a unique number and name.

24. Select the paradigm you created from the Paradigm Name pull-down list.

25. Complete all the fields on the Paradigm Manager window. In this visual paradigm example, enter the following values.

- TR = 4000
- Group Delay = 0
- Dummy Volumes = 4
- Initial State = 0
- Number of Rest Volumes = 6 (24 seconds/4000ms = 6).
- Number of Stimulus Volumes = 6
- Number of Phases = 96 (8 rest and 8 stimulus cycles = 16. 16 cycles x 6 volumes = 96).
- Paradigm Payout Time = 6 minutes, 24 seconds (24 seconds per epoch x 8 = 192 seconds. 192 rest and 192 task = 384 seconds. 384/60 = 6.4 minutes or 6 minutes and 24 seconds).

Paradigm Manager	
Hardware Paradigms Software Paradigms Settings/Help	
Paradigm Name	4000 6 6 96 vizchecker
TR (msec)	4000
Group Delay (msec)	0
Dummy Volumes	4
Initial State	0
Number of Rest Volumes	6
Number of Stimulus Volumes	6
Number of Phases	96
Paradigm Payout Time	06:24.000
Scan Time	06:40.000

26. Click **Apply**.

- Apply allows you to change the parameters on the Hardware Paradigms tab until the **[Save]** button is clicked.

27. Click **Save** to save your new paradigm.

- Once the paradigm is saved, if you want to change one of the scan parameters associated with the paradigm, you must delete the paradigm, refresh the list, re-enter the desired paradigm, apply the parameters, and save the paradigm again.
- If you truly want to delete a Hardware paradigm, you have to delete the item within the Paradigm Manager and delete the .stm file from the Stimulus PC (using Windows Explorer through virtually connecting to the Stimulus PC).

28. Verify the new paradigm appears in BrainWaveRT.

- Return to fMRI Additional Parameters in BrainWaveRT.
- Select the Hardware paradigm list and check for the newly created paradigm name.

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